Osdag
Open steel design and graphics

OsdagBridge

Open Source Software for Steel Girder Bridge Design

Design Report

Project Name	Sample Plate Girder Bridge
Project Location	Pune, Maharashtra
Author / Designer	Vivek Manik Warkad
Reviewer	Osdag Team
Organization	IIT Kanpur
Client Name and Organization	FOSSEE, IIT Bombay
Job Number	OB-2026-001
Date	2026-11-13
Report Version	Rev 0

Table of Contents

Executive Summary **4**

Project Overview 4

Key Design Outcomes Summary 6

Design Assumptions and Limitations 6

1 Project Information **7**

1.1 Project and Design Team Details 7

1.2 Applicable Codes and Standards 7

2 Input Parameters **8**

2.1 Basic Inputs (User-Defined) 8

2.2 Additional Inputs 9

3 Loads and Load Combinations **13**

3.1 Permanent Loads 13

3.2 Live Loads 13

3.3 Environmental Loads 15

3.4 Load Combinations 16

4 Analysis Results **18**

4.1 Maximum Demands 18

4.2 Support Reactions 20

4.3 Deflections 22

4.4 Analysis Diagrams 23

5 Design Checks **25**

5.1 Plate Girder Design 25

5.2 Deck Slab Design 38

5.3 Cross Bracing Design 41

5.4 End Diaphragm Design 43

5.5 Overall Design Check Summary 45

6 Drawings and Visualizations **47**

6.1 Bridge Configuration and Layout 47

6.2 Plate Girder — Detailed Views 49

6.3 Cross Bracing Detail 50

6.4 End Diaphragm Detail 51

7 Material Take-off & Quantity Summary **53**

7.1 Bill of Materials 53

7.2 Material Quantity Charts 54

8 Standards & Assumptions **56**

8.1 Design Standards 56

8.2 Analysis and Design Assumptions of This Version 57

8.3 Known Limitations of This Version 59

References **60**

Executive Summary

This section provides a concise summary of the bridge design, key inputs, governing loads, and final design outcomes.

Project Overview

Table ES.1: Project Overview

Parameter	Value
Bridge Type	Highway Bridge
Design Standards	IRC 5, IRC 6, IRC 22, IRC 24, IS 800
Span	30 m
Carriageway Width	10.5 m
No. of Girders	5
Girder Spacing	2.28 m
Deck Thickness	250 mm
Overall Design Status (girders)	**FAIL** — Fatigue Normal Stress, Int.Stiff: Leg ≤ H_ limit, Fatigue Shear Stress
Governing Girder Check	Fatigue Normal Stress
Overall Utilization Ratio (max.)	**1.87** (Steel Plate Girders)

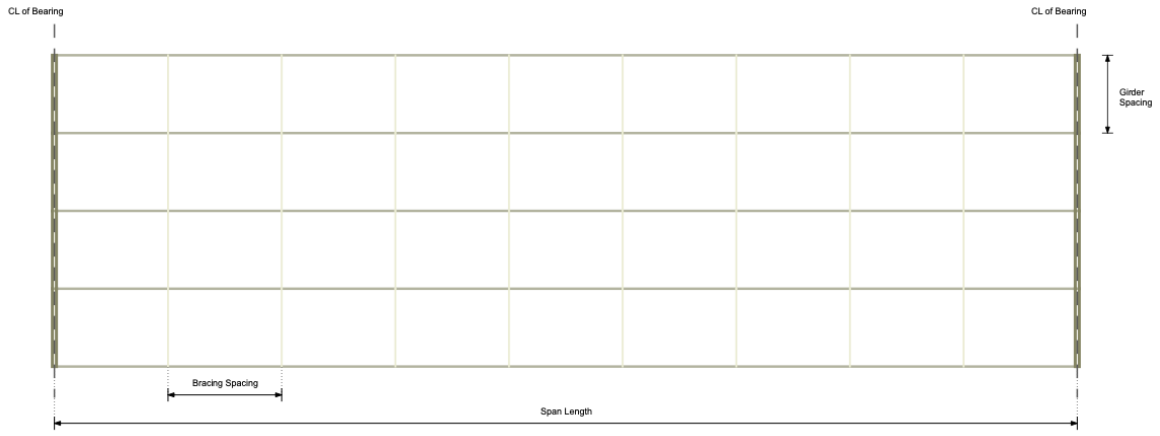


Figure ES.1: Overall bridge plan

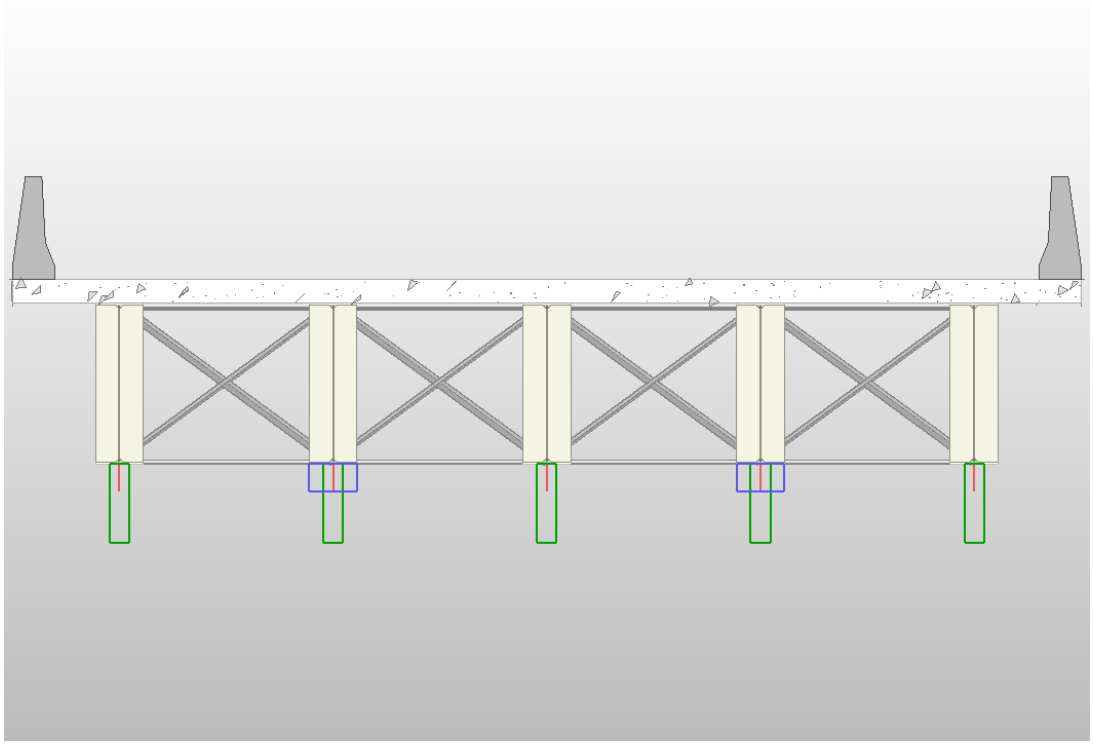


Figure ES.2: Typical cross-section (girders, deck, barriers, footpath)

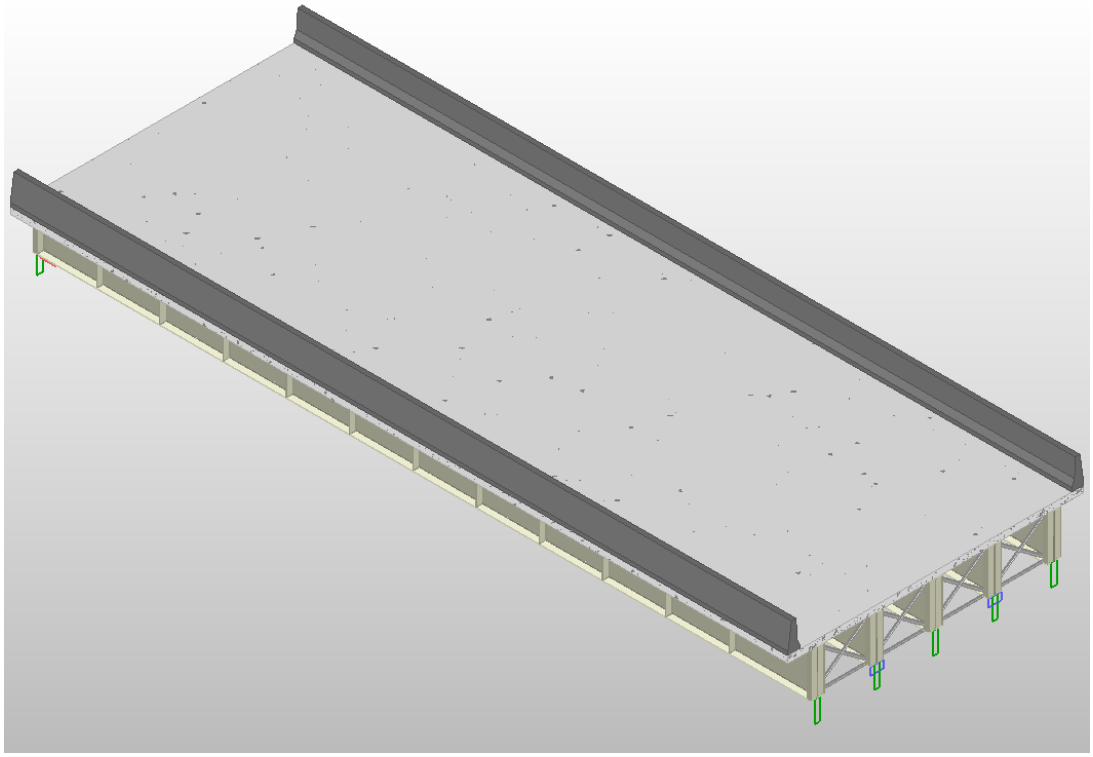


Figure ES.3: 3D view of the bridge superstructure

Table ES.2: Final Girder Sections (after optimization)

Girder	Member ID	Section Designation ($D \times b_{tf} \times t_{tf} \times b_{bf} \times t_{bf}$, mm)	Governing Check	UR	Status
G1	G1M1	1670 x 510 x 22 x 510 x 22	Fatigue Normal Stress	**1.80**	**FAIL**
G2	G2M1	1670 x 510 x 22 x 510 x 22	Fatigue Normal Stress	**1.81**	**FAIL**
G3	G3M1	1670 x 510 x 22 x 510 x 22	Fatigue Normal Stress	**1.83**	**FAIL**
G4	G4M1	1670 x 510 x 22 x 510 x 22	Fatigue Normal Stress	**1.86**	**FAIL**
G5	G5M1	1670 x 510 x 22 x 510 x 22	Fatigue Normal Stress	**1.87**	**FAIL**

Utilization ratio (UR) = demand / capacity. A value ≤ 1.0 indicates a passing check.

Key Design Outcomes Summary

Table ES.3: Key Design Outcomes by Structural Element

Structural Element	Governing Check	Utilization Ratio	Status
Steel Plate Girders	Fatigue	**1.87**	**FAIL**
Concrete Deck Slab	Cantilever overhang	0.91	**PASS**
Cross Bracing	Compression	0.93	**PASS**
End Diaphragms	Compression	0.93	**PASS**

Details of every check are given in Chapter 5; Section 5.5 shows the utilization-ratio charts.

Design Assumptions and Limitations

- Additional inputs not provided by the user were assumed by software per IRC/IS code defaults or practical consideration.
- Grillage analysis was performed using OSPGrillage assuming simply supported I-girders.
- Substructure and foundation design are not included in this report.
- Splice connections and bearings are not designed in this version.

1 Project Information

This section records all project metadata as entered by the designer.

1.1 Project and Design Team Details

Table 1.1: Project and Design Team Details

Item	Details
Project Name	Sample Plate Girder Bridge
Project Location	Pune, Maharashtra
Designer	Vivek Manik Warkad
Reviewer	Osdag Team
Organization	IIT Kanpur
Client	FOSSEE, IIT Bombay
Job Number	OB-2026-001
Report Date	2026-11-13
Software	OsdagBridge

1.2 Applicable Codes and Standards

- Indian Roads Congress (IRC) 5: General Features of Design
- Indian Roads Congress (IRC) 6: Loads and Load Combinations
- Indian Roads Congress (IRC) 22: Composite Construction (Limit State Design)
- Indian Roads Congress (IRC) 24: Steel Road Bridges (Limit State Method)
- Indian Roads Congress (IRC) 112: Concrete Road Bridges (deck design)
- Indian Roads Congress Special Publication (IRC SP) 114: Seismic Design of Road Bridges
- Indian Standard (IS) 800: General Construction in Steel
- Indian Standard (IS) 2062: Hot Rolled Structural Steel Specification
- Indian Standard (IS) 6006: Steel Bearings

2 Input Parameters

This section documents all inputs provided to OsdagBridge. User-provided inputs are clearly distinguished from software-assumed defaults. Where the user did not supply a value, the software has applied the IRC/IS code default or an empirical guideline; these are annotated with an asterisk (*).

2.1 Basic Inputs (User-Defined)

These inputs are mandatory and were provided by the user.

Table 2.1: Project Location

Parameter	Value	Unit	Reference / Remarks
Project Location	Pune, Maharashtra	—	Basic Inputs
Latitude / Longitude	18.5204, 73.8567	°	From project location
Seismic Zone	III	—	IRC 6 Fig. 18
Basic Wind Speed	39	m/s	IRC 6 Fig. 10
Shade Air Temperature — Maximum	43.3	°C	IRC 6 Fig. 15
Shade Air Temperature — Minimum	1.7	°C	IRC 6 Fig. 16

Table 2.2: Bridge Geometry

Parameter	Value	Unit	Reference / Remarks
Type of Structure	Highway Bridge	—	Basic Inputs
Span	30	m	Effective span, simply supported
Carriageway Width	10.5	m	IRC 5 Cl. 104.3.1
Include Median	No	—	Basic Inputs
Footpath	None	—	Basic Inputs
Skew Angle	—	°	IRC 24 Cl. 504.8 limit: ±15°

** Software default value*

Table 2.3: Material Selection

Parameter	Value
Girder Steel Grade (IS 2062)	E 350A
Cross Bracing Steel Grade	E 350A
End Diaphragm Steel Grade	E 350A
Concrete Deck Grade (IRC 22)	M40

2.2 Additional Inputs

Where the user has modified additional inputs, those values are reported here. Where no modification was made, the software default is shown.

Table 2.4: Typical Section Details

Parameter	Value	Unit	Reference / Remarks
Overall Bridge Width	11.4	m	Carriageway + barriers + footpaths
No. of Girders	5	—	Additional Inputs
Girder Spacing	2.28	m	Centre to centre
Deck Overhang Width	1.14	m	Beyond the outer girder
Deck Thickness	250	mm	Additional Inputs
Footpath Width	0	m	IRC 5 Cl. 104.3.6 (min. 1.5 m)
No. of Traffic Lanes	—	—	IRC 5 Cl. 104.3.1

Table 2.5: Components Details

Parameter	Value	Unit	Reference / Remarks
Crash Barrier Type	IRC 5 - RCC Crash Barrier	—	IRC 5 Cl. 109
Crash Barrier Load	7.557	kN/m	Per barrier
Railing Type	IRC 5 - RCC Railing	—	Additional Inputs
Railing Load	1.4715	kN/m	IRC 6 Cl. 206.5
Wearing Course Material	Concrete	—	Additional Inputs
Wearing Course Thickness	50	mm	Additional Inputs

Table 2.6: Girder General Information

Girder	Member ID	Design Mode	Girder Type	Girder Symmetry
G1	G1M1	Optimized	Welded	Girder Symmetric
G2	G2M1	Optimized	Welded	Girder Symmetric
G3	G3M1	Optimized	Welded	Girder Symmetric
G4	G4M1	Optimized	Welded	Girder Symmetric
G5	G5M1	Optimized	Welded	Girder Symmetric

Table 2.7: Girder Section Dimensions

Girder	Total Depth, D (mm)	Web, t_w (mm)	Top Flange, $b_{tf} \times t_{tf}$ (mm)	Bottom Flange, $b_{bf} \times t_{bf}$ (mm)
G1	1670	10	510 × 22	510 × 22
G2	1670	10	510 × 22	510 × 22
G3	1670	10	510 × 22	510 × 22
G4	1670	10	510 × 22	510 × 22
G5	1670	10	510 × 22	510 × 22

Table 2.8: Girder Restraint and Stiffener Details

Girder	Torsional / Warping Restraint	Web Philosophy	Intermediate Stiffeners	Longitudinal Stiffeners	Bearing Stiffeners
G1	Fully Restrained; Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G2	Fully Restrained; Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G3	Fully Restrained; Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm

Continued on next page

Table 2.8: Girder Restraint and Stiffener Details (continued)

Girder	Torsional / Warping Restraint	Web Philosophy	Intermediate Stiffeners	Longitudinal Stiffeners	Bearing Stiffeners
G4	Fully Restrained; Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm
G5	Fully Restrained; Both Flanges Restrained	Thin Web with ITS	Yes; Spacing: 2440 mm; Thickness: 10 mm	No	No.: 4; Spacing: 100 mm; Thickness: 10 mm

Table 2.9: Member Properties: Cross Bracing Details

Location	Member IDs	Type of Bracing	Bracing Section	Spacing (m)
G1 to G2	B1M1 to B1M1	X-Bracing	IS 100 x 100 x 10	15
G2 to G3	B2M1 to B2M1	X-Bracing	IS 100 x 100 x 10	15
G3 to G4	B3M1 to B3M1	X-Bracing	IS 100 x 100 x 10	15
G4 to G5	B4M1 to B4M1	X-Bracing	IS 100 x 100 x 10	15

Table 2.10: Member Properties: End Diaphragm Details

Location	Member IDs	Type of Diaphragm	Section
G1 to G2	E1M1 to E1M2	Cross Bracing	IS 100 x 100 x 10
G2 to G3	E2M1 to E2M2	Cross Bracing	IS 100 x 100 x 10
G3 to G4	E3M1 to E3M2	Cross Bracing	IS 100 x 100 x 10
G4 to G5	E4M1 to E4M2	Cross Bracing	IS 100 x 100 x 10

Table 2.11: Shear Connector Details

Parameter	Value	Unit	Reference / Remarks
Stud Diameter	20	mm	IRC 22 Cl. 606.6
Stud Height	100	mm	IRC 22 Cl. 606.6
Stud Yield Strength, f_y	385	MPa	IRC 22 Cl. 606.3
Stud Ultimate Strength, f_u	495	MPa	IRC 22 Cl. 606.3
No. of Studs per Section	2	—	Across the flange width

Table 2.12: Partial Safety Factors

Partial Safety Factor	Value
γ_{M0} (Yielding / Buckling)	1.1
γ_{M1} (Ultimate Stress)	1.25
γ_C (Concrete, Basic)	1.5
γ_s (Reinforcement)	1.15
γ_v (Shear Connectors)	1.25
γ_{fft} (Fatigue Load)	1
γ_{Mft} (Fatigue Strength)	1.35

All values are per IRC 22 Table 1 unless user-modified.

3 Loads and Load Combinations

This section summarizes all loads applied to the bridge and the load combinations considered for analysis and design.

3.1 Permanent Loads

Table 3.1: Dead Load — Self Weight (DL)

Parameter	Value	Unit	Reference / Remarks
Unit Weight of Structural Steel	76.4	kN/m^3	IRC 6 Cl. 203
Unit Weight of Deck Concrete	24.5	kN/m^3	IRC 6 Cl. 203
Self-Weight Factor	1	—	Multiplier on the computed self weight

Table 3.2: Superimposed Dead Load — Surfacing and Barriers (DW)

Parameter	Value	Unit	Reference / Remarks
Wearing Course Material	Concrete	—	Additional Inputs
Wearing Course Thickness	50	mm	Additional Inputs
Crash Barrier Load	7.557	kN/m	Per barrier (SIDL)
Railing Load	1.4715*	kN/m	IRC 6 Cl. 206.5

3.2 Live Loads

Vehicular live loads and the footpath (pedestrian) live load are reported separately. Only the IRC vehicles selected by the user in the Loading tab of the Additional Inputs are listed.

** Software default value*

Table 3.3: Vehicle Live Loads (LL) — IRC 6

Vehicle	Total Load (kN)	Impact Factor	Braking Load		
			Considered?	Value (kN)	Eccentricity (m)
Class A	543.5	1.207	Yes	108.7	1.20
Class AA Wheeled	392.4	1.207	Yes	78.5	1.20
Class AA Tracked	686.7	1.100	Yes	137.3	1.20
Class 70R Wheeled	981.0	1.207	Yes	196.2	1.20
Class 70R Tracked	686.7	1.100	Yes	137.3	1.20
Class 70R Bogie	392.4	1.207	Yes	78.5	1.20
Class SV	3776.9	1.000	Yes	755.4	1.20
Class Fatigue	392.4	1.207	Yes	78.5	1.20

Vehicles are those selected under Additional Inputs → Loading → Live Load. Total load is that of one vehicle / train (IRC 6 Cl. 204). Impact factor = $1 + I$ for steel bridges (IRC 6 Cl. 208): Class A and fatigue vehicle $I = 9/(13.5 + L)$; tracked vehicles $I = 0.10$; wheeled Class AA / 70R $I = 0.25$ up to 23 m span, $9/(13.5 + L)$ beyond; no impact for Class SV (Cl. 204.5). Braking value = 20% of the vehicle load (first train, IRC 6 Cl. 211.2), acting at the stated eccentricity above the top of the deck.

Table 3.4: Live Load — Braking and Centrifugal Forces

Parameter	Value	Unit	Reference / Remarks
No. of Design Traffic Lanes	—	—	IRC 6 Table 6
Braking Force — Governing Vehicle (Class SV)	755.4	kN	20% of first train, IRC 6 Cl. 211.2(a)
Braking Force — Lanes in Excess of Two	N/A	kN	5% of the load on those lanes, IRC 6 Cl. 211.2(b)
Design Braking Force (longitudinal)	755.4	kN	Applied along the roadway
Braking Force — Height above Deck	1.20	m	IRC 6 Cl. 211.3
Centrifugal Force	N/A	kN	IRC 6 Cl. 212: $C = WV^2/(127R)$; not applicable to a straight alignment

Table 3.5: Footpath Live Load — IRC 6

Parameter	Value	Unit	Reference / Remarks
Footpath Provided	None	—	Basic Inputs
Footpath Live Load	N/A	kN/m^2	No footpath on this bridge

3.3 Environmental Loads

Table 3.6: Wind Load (WL) — IRC 6

Parameter	Value	Unit	Reference / Remarks
Basic Wind Speed, V_b	39	m/s	From project location
Terrain Type	Plain Terrain	—	IRC 6 Cl. 209.2
Average Exposed Height, H	10	m	Additional Inputs
Hourly Mean Wind Speed, V_z	32.85	m/s	IRC 6 Table 12
Hourly Wind Pressure, P_z	647.65	N/m^2	IRC 6 Table 12
Transverse Wind Force	—	kN	IRC 6 Cl. 209.3.3
Longitudinal Wind Force	—	kN	IRC 6 Cl. 209.3.4
Vertical Wind Force	—	kN	IRC 6 Cl. 209.3.5

Table 3.7: Earthquake Load (EL) — IRC 6

Parameter	Value	Unit	Reference / Remarks
Seismic Zone	III	—	From project location
Zone Factor, Z	0.16	—	IRC 6 Table 16
Importance Factor, I	1	—	IRC 6 Table 19
Type of Soil	Type I - Rocky or Hard	—	IRC 6 Cl. 218.5
Spectral Acceleration Coefficient, S_a/g	2.8	—	IRC 6 Cl. 218.5.1
Horizontal Seismic Coefficient, A_h	0.2240	—	IRC 6 Cl. 218.5.1
Vertical Seismic Coefficient, A_v	0.1493	—	$\frac{2}{3}A_h$, IRC 6 Cl. 218.3

Table 3.8: Temperature Load (TL) — IRC 6

Parameter	Value	Unit	Reference / Remarks
Maximum Shade Air Temperature	43.3	°C	From project location
Minimum Shade Air Temperature	1.7	°C	From project location
Effective Bridge Temperature — Minimum	-8.30	°C	IRC 6 Cl. 215.2
Effective Bridge Temperature — Maximum	58.30	°C	IRC 6 Cl. 215.2
Design Temperature Rise	+33.30	°C	From mean bridge temperature
Design Temperature Fall	−33.30	°C	From mean bridge temperature

3.4 Load Combinations

Table 3.9: Load Combinations

ID	Description	Load Cases and Partial Factors
ULS-01	Basic - LL leading	$(1.35/1.00)\,\mathrm{DL} + (1.75/1.00)\,\mathrm{SIDL} + 1.50\,\mathrm{LL} + 0.90\,\mathrm{WL} + 0.90\,\mathrm{TL}$
ULS-02	Basic - WL leading	$(1.35/1.00)\,\mathrm{DL} + (1.75/1.00)\,\mathrm{SIDL} + 1.15\,\mathrm{LL} + 1.50\,\mathrm{WL} + 0.90\,\mathrm{TL}$
ULS-03	Basic - TL leading	$(1.35/1.00)\,\mathrm{DL} + (1.75/1.00)\,\mathrm{SIDL} + 1.15\,\mathrm{LL} + 0.90\,\mathrm{WL} + 1.50\,\mathrm{TL}$
ULS-04	Accidental (VC) - LL leading	$(1.00/1.00)\,\mathrm{DL} + (1.00/1.00)\,\mathrm{SIDL} + 0.75\,\mathrm{LL} + 0.50\,\mathrm{TL} + 1.00\,\mathrm{VC}$
ULS-05	Accidental (BI) - LL leading	$(1.00/1.00)\,\mathrm{DL} + (1.00/1.00)\,\mathrm{SIDL} + 0.75\,\mathrm{LL} + 0.50\,\mathrm{TL} + 1.00\,\mathrm{BI}$
ULS-06	Accidental (FB) - LL leading	$(1.00/1.00)\,\mathrm{DL} + (1.00/1.00)\,\mathrm{SIDL} + 0.75\,\mathrm{LL} + 0.50\,\mathrm{TL} + 1.00\,\mathrm{FB}$
ULS-07	Seismic (service) - all variable loads accompanying	$(1.35/1.00)\,\mathrm{DL} + (1.75/1.00)\,\mathrm{SIDL} + 0.20\,\mathrm{LL} + 0.50\,\mathrm{TL} + 1.50\,\mathrm{EQ}$
ULS-08	Seismic (construction) - all variable loads accompanying	$(1.35/1.00)\,\mathrm{DL} + (1.75/1.00)\,\mathrm{SIDL} + 0.20\,\mathrm{LL} + 0.50\,\mathrm{TL} + 0.75\,\mathrm{EQ}$
SLS-01	SLS Rare - LL leading	$1.00\,\mathrm{DL} + (1.20/1.00)\,\mathrm{SIDL} + 1.00\,\mathrm{LL} + 0.60\,\mathrm{WL} + 0.60\,\mathrm{TL}$
SLS-02	SLS Rare - WL leading	$1.00\,\mathrm{DL} + (1.20/1.00)\,\mathrm{SIDL} + 0.75\,\mathrm{LL} + 1.00\,\mathrm{WL} + 0.60\,\mathrm{TL}$
SLS-03	SLS Rare - TL leading	$1.00\,\mathrm{DL} + (1.20/1.00)\,\mathrm{SIDL} + 0.75\,\mathrm{LL} + 0.60\,\mathrm{WL} + 1.00\,\mathrm{TL}$
SLS-04	SLS Frequent - LL leading	$1.00\,\mathrm{DL} + (1.20/1.00)\,\mathrm{SIDL} + 0.75\,\mathrm{LL} + 0.50\,\mathrm{WL} + 0.50\,\mathrm{TL}$
SLS-05	SLS Frequent - WL leading	$1.00\,\mathrm{DL} + (1.20/1.00)\,\mathrm{SIDL} + 0.20\,\mathrm{LL} + 0.60\,\mathrm{WL} + 0.50\,\mathrm{TL}$

Continued on next page

Table 3.9: Load Combinations (continued)

ID	Description	Load Cases and Partial Factors
SLS-06	SLS Frequent - TL leading	$1.00\,\mathrm{DL} + (1.20/1.00)\,\mathrm{SIDL} + 0.20\,\mathrm{LL} + 0.50\,\mathrm{WL} + 0.60\,\mathrm{TL}$
SLS-07	SLS Quasi-permanent - all variable loads accompanying	$1.00\,\mathrm{DL} + (1.20/1.00)\,\mathrm{SIDL} + 0.00\,\mathrm{LL} + 0.00\,\mathrm{WL} + 0.50\,\mathrm{TL}$

All IRC 6 load combinations are auto-generated by OsdagBridge. For permanent loads the two factors are (adding / relieving) effect. User-defined custom combinations, if any, are appended.

4 Analysis Results

A grillage model was used for structural analysis. The deck is idealized as a grid of elastic beam elements — longitudinal members represent the composite steel girders with effective slab, and transverse members represent the slab or cross frames. This section summarizes the critical output from that analysis.

4.1 Maximum Demands

Table 4.1: Summary of Maximum Demands

Load Case / Combination	Bending Moment			Shear Force		
	Max (kN-m)	Girder	Location (m)	Max (kN)	Girder	Location (m)
Basic load cases						
1.0 DL	1378.46	G1	15.00	236.42	G1	28.50
1.0 DL + 1.0 LL	2427.03	G4	13.50	434.20	G1	0.00
1.0 DW	222.35	G1	15.00	38.28	G1	0.00
1.0 WL	128.62	G1	15.00	21.76	G1	0.00
1.5 EQ (a)	107.37	G1	15.00	18.17	G1	0.00
1.5 EQ (b)	107.37	G1	15.00	18.17	G1	28.50
1.5 EQ (c)	357.89	G1	15.00	60.56	G1	28.50
Crash barrier load	139.67	G3	15.00	26.36	G1	28.50
DD	1158.06	G1	15.00	199.38	G1	28.50
EQ_X	0.00	—	—	0.00	—	—
EQ_Y	238.59	G1	15.00	40.37	G1	28.50
EQ_Z	0.00	—	—	0.00	—	—
SW	220.41	G1	15.00	37.04	G1	28.50
WL Longitudinal	0.00	—	—	0.00	—	—
WL Transverse	0.00	—	—	0.00	—	—
WL Uplift	128.62	G1	15.00	21.76	G1	28.50
Load combinations (IRC 6)						
ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	2646.84	G4	13.50	472.49	G1	0.00
ACCIDENTAL_2: 1.0DL + 1.0DW + 0.75LL	2646.84	G4	13.50	472.49	G1	0.00
ACCIDENTAL_3: 1.0DL + 1.0DW + 0.75LL	2646.84	G4	13.50	472.49	G1	0.00

Continued on next page

Table 4.1: Summary of Maximum Demands (continued)

Load Case / Combination	Bending Moment			Shear Force		
	Max (kN-m)	Girder	Location (m)	Max (kN)	Girder	Location (m)
BASIC_1: 1.35DL + 1.75DW + 1.5LL + 0.9WL	2519.69	G4	13.50	450.72	G1	0.00
BASIC_2: 1.35DL + 1.75DW + 1.15LL + 1.5WL	2519.69	G4	13.50	450.72	G1	0.00
BASIC_3: 1.35DL + 1.75DW + 1.15LL + 0.9WL	2519.69	G4	13.50	450.72	G1	0.00
BASIC_4: 1.0DL + 1.0DW + 1.5LL + 0.9WL	2519.69	G4	13.50	450.72	G1	0.00
BASIC_5: 1.0DL + 1.0DW + 1.15LL + 1.5WL	2519.69	G4	13.50	450.72	G1	0.00
BASIC_6: 1.0DL + 1.0DW + 1.15LL + 0.9WL	2519.69	G4	13.50	450.72	G1	0.00
SEISMIC_1: 1.35DL + 1.75DW + 0.2LL + 1.5EL	2410.99	G4	13.50	432.11	G1	0.00
SEISMIC_2: 1.35DL + 1.75DW + 0.2LL + 0.75EL	2410.99	G4	13.50	432.11	G1	0.00
SEISMIC_3: 1.0DL + 1.0DW + 0.2LL + 1.5EL	2410.99	G4	13.50	432.11	G1	0.00
SEISMIC_4: 1.0DL + 1.0DW + 0.2LL + 0.75EL	2410.99	G4	13.50	432.11	G1	0.00
SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	2519.69	G4	13.50	450.72	G1	0.00
SLS_FREQUENT_2: 1.0DL + 1.2DW + 0.2LL + 0.6WL	2519.69	G4	13.50	450.72	G1	0.00
SLS_FREQUENT_3: 1.0DL + 1.2DW + 0.2LL + 0.5WL	2519.69	G4	13.50	450.72	G1	0.00
SLS_FREQUENT_4: 1.0DL + 1.0DW + 0.75LL + 0.5WL	2519.69	G4	13.50	450.72	G1	0.00
SLS_FREQUENT_5: 1.0DL + 1.0DW + 0.2LL + 0.6WL	2519.69	G4	13.50	450.72	G1	0.00
SLS_FREQUENT_6: 1.0DL + 1.0DW + 0.2LL + 0.5WL	2519.69	G4	13.50	450.72	G1	0.00
SLS_QP_1: 1.0DL + 1.2DW	1600.81	G1	15.00	274.70	G1	0.00
SLS_QP_2: 1.0DL + 1.0DW	1600.81	G1	15.00	274.70	G1	28.50
SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	2519.69	G4	13.50	450.72	G1	0.00
SLS_RARE_2: 1.0DL + 1.2DW + 0.75LL + 1.0WL	2519.69	G4	13.50	450.72	G1	0.00
SLS_RARE_3: 1.0DL + 1.2DW + 0.75LL + 0.6WL	2519.69	G4	13.50	450.72	G1	0.00

Continued on next page

Table 4.1: Summary of Maximum Demands (continued)

Load Case / Combination	Bending Moment			Shear Force		
	Max (kN-m)	Girder	Location (m)	Max (kN)	Girder	Location (m)
SLS_RARE_4: 1.0DL + 1.0DW + 1.0LL + 0.6WL	2519.69	G4	13.50	450.72	G1	0.00
SLS_RARE_5: 1.0DL + 1.0DW + 0.75LL + 1.0WL	2519.69	G4	13.50	450.72	G1	0.00
SLS_RARE_6: 1.0DL + 1.0DW + 0.75LL + 0.6WL	2519.69	G4	13.50	450.72	G1	0.00
Envelopes						
Envelope SLS	2519.69	G4	13.50	450.72	G1	0.00
Envelope ULS	2646.84	G4	13.50	472.49	G1	0.00

Location is measured from the left support along the span.

4.2 Support Reactions

Table 4.2: Reactions at Supports

Load Case / Combination	Left Support (kN)	Right Support (kN)
Basic load cases		
1.0 DL	-990.41	990.41
1.0 DL + 1.0 LL	-1788.51	1364.29
1.0 DW	-159.94	159.94
1.0 WL	91.74	-91.74
1.5 EQ (a)	76.58	-76.58
1.5 EQ (b)	76.58	-76.58
1.5 EQ (c)	255.26	-255.26
Crash barrier load	-91.69	96.53
DD	-833.01	833.01
EQ_X	0.00	0.00
EQ_Y	170.18	-170.18
EQ_Z	0.00	0.00
SW	-157.41	157.41
WL Longitudinal	0.00	0.00
WL Transverse	0.00	0.00
WL Uplift	91.74	-91.74

Continued on next page

Table 4.2: Reactions at Supports (continued)

Load Case / Combination	Left Support (kN)	Right Support (kN)
Load combinations (IRC 6)		
ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	-1948.44	1524.23
ACCIDENTAL_2: 1.0DL + 1.0DW + 0.75LL	-1948.44	1524.23
ACCIDENTAL_3: 1.0DL + 1.0DW + 0.75LL	-1948.44	1524.23
BASIC_1: 1.35DL + 1.75DW + 1.5LL + 0.9WL	-1856.70	1432.49
BASIC_2: 1.35DL + 1.75DW + 1.15LL + 1.5WL	-1856.70	1432.49
BASIC_3: 1.35DL + 1.75DW + 1.15LL + 0.9WL	-1856.70	1432.49
BASIC_4: 1.0DL + 1.0DW + 1.5LL + 0.9WL	-1856.70	1432.49
BASIC_5: 1.0DL + 1.0DW + 1.15LL + 1.5WL	-1856.70	1432.49
BASIC_6: 1.0DL + 1.0DW + 1.15LL + 0.9WL	-1856.70	1432.49
SEISMIC_1: 1.35DL + 1.75DW + 0.2LL + 1.5EL	-1778.27	1354.05
SEISMIC_2: 1.35DL + 1.75DW + 0.2LL + 0.75EL	-1778.27	1354.05
SEISMIC_3: 1.0DL + 1.0DW + 0.2LL + 1.5EL	-1778.27	1354.05
SEISMIC_4: 1.0DL + 1.0DW + 0.2LL + 0.75EL	-1778.27	1354.05
SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	-1856.70	1432.49
SLS_FREQUENT_2: 1.0DL + 1.2DW + 0.2LL + 0.6WL	-1856.70	1432.49
SLS_FREQUENT_3: 1.0DL + 1.2DW + 0.2LL + 0.5WL	-1856.70	1432.49
SLS_FREQUENT_4: 1.0DL + 1.0DW + 0.75LL + 0.5WL	-1856.70	1432.49
SLS_FREQUENT_5: 1.0DL + 1.0DW + 0.2LL + 0.6WL	-1856.70	1432.49
SLS_FREQUENT_6: 1.0DL + 1.0DW + 0.2LL + 0.5WL	-1856.70	1432.49
SLS_QP_1: 1.0DL + 1.2DW	-1150.35	1150.35

Continued on next page

Table 4.2: Reactions at Supports (continued)

Load Case / Combination	Left Support (kN)	Right Support (kN)
SLS_QP_2: 1.0DL + 1.0DW	-1150.35	1150.35
SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	-1856.70	1432.49
SLS_RARE_2: 1.0DL + 1.2DW + 0.75LL + 1.0WL	-1856.70	1432.49
SLS_RARE_3: 1.0DL + 1.2DW + 0.75LL + 0.6WL	-1856.70	1432.49
SLS_RARE_4: 1.0DL + 1.0DW + 1.0LL + 0.6WL	-1856.70	1432.49
SLS_RARE_5: 1.0DL + 1.0DW + 0.75LL + 1.0WL	-1856.70	1432.49
SLS_RARE_6: 1.0DL + 1.0DW + 0.75LL + 0.6WL	-1856.70	1432.49
Envelopes		
Envelope SLS	-1856.70	1432.49
Envelope ULS	-1948.44	1524.23

Total vertical reaction of each support line, in the sign convention of the grillage model.

4.3 Deflections

Table 4.3: Deflection Summary (Live Load and Total Load)

Deflection	Computed (mm)	Allowable (mm)	Status
Due to Live Load, δ_{LL}	25.150	L/800 = 37.5	**PASS**
Due to Total Load, δ_{total}	61.738	L/600 = 50.0	**FAIL**

Governing (largest) deflection of all girders; limits per IRC 22 Cl. 604.3.2.

4.4 Analysis Diagrams

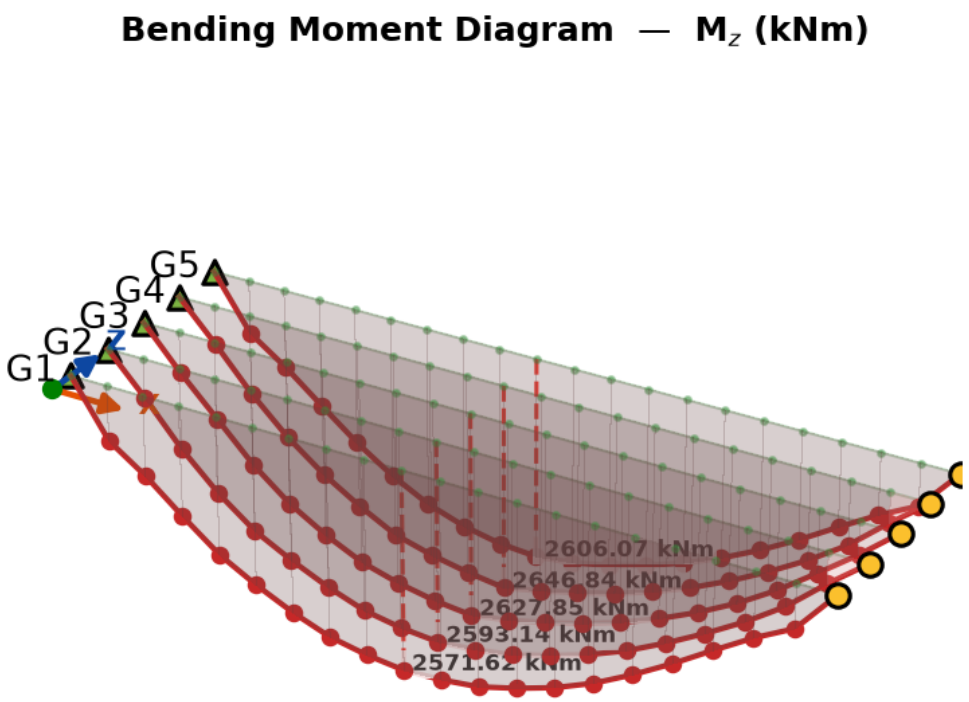


Figure 4.1: Bending moment envelope (Envelope ULS): maximum / minimum bending moment (kN-m) along the span, measured from the left support (m)

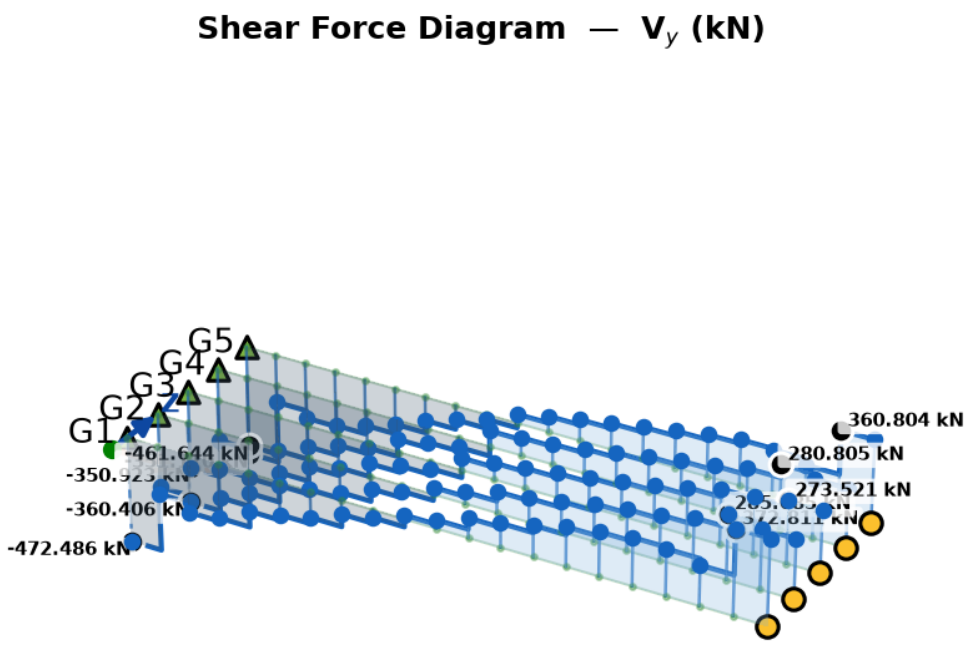


Figure 4.2: Shear force envelope (Envelope ULS): maximum / minimum shear force (kN) along the span, measured from the left support (m)

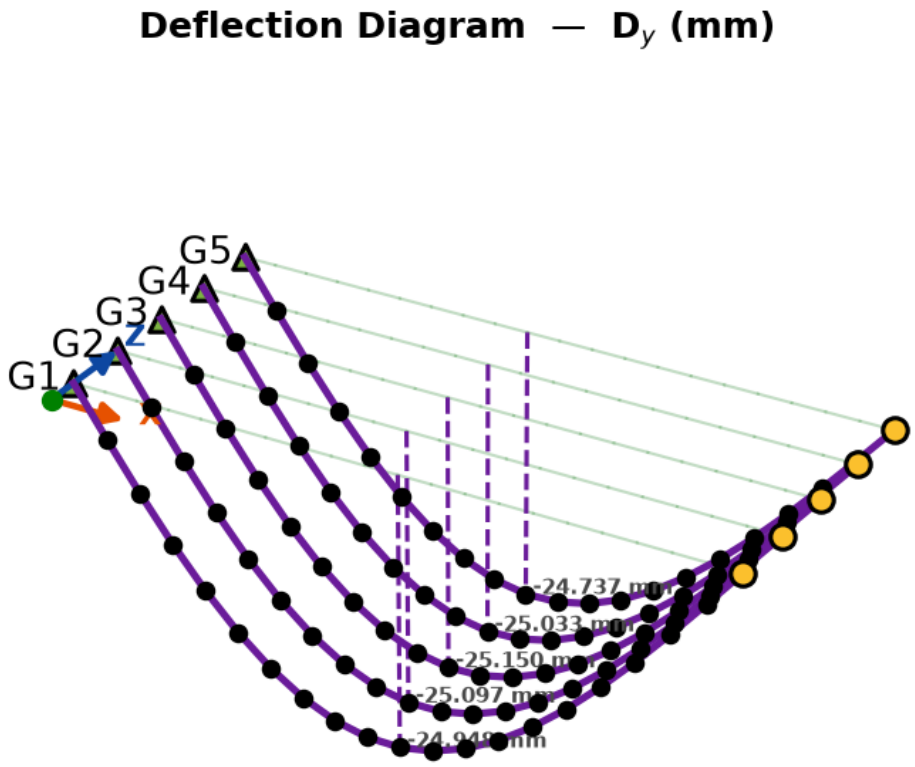


Figure 4.3: Vertical deflection D_y under live load (1.0 LL): deflected shape of the grillage, isometric view

5 Design Checks

This section presents all structural design checks performed by OsdagBridge. For each member, the demand from the governing load combination, the code-based capacity, and the utilization ratio are tabulated. All checks reference IS 800:2007 and IRC 22:2014 unless stated otherwise.

5.1 Plate Girder Design

Table 5.1: Girder Section Properties (Final Optimized / User-selected)

Girder	Property	Value	Unit
G1	Depth, D	1670	mm
	Top Flange Width, b_f	510	mm
	Bottom Flange Width, b_f	510	mm
	Top Flange Thickness, t_f	22	mm
	Bottom Flange Thickness, t_f	22	mm
	Web Thickness, t_w	10	mm
	Gross Area, A	387	cm^2
	Moment of Inertia, I_z	1881957.85	cm^4
	Elastic Section Modulus, Z_{ez}	22538.42	cm^3
	Plastic Section Modulus, Z_{pz}	25100.25	cm^3
	Effective Slab Width, b_{eff}	2280	mm
	Transformed Composite I_z	4284758.2	cm^4
	Depth to Plastic Neutral Axis	262.8	mm
G2	Depth, D	1670	mm
	Top Flange Width, b_f	510	mm
	Bottom Flange Width, b_f	510	mm
	Top Flange Thickness, t_f	22	mm
	Bottom Flange Thickness, t_f	22	mm
	Web Thickness, t_w	10	mm
	Gross Area, A	387	cm^2
	Moment of Inertia, I_z	1881957.85	cm^4
	Elastic Section Modulus, Z_{ez}	22538.42	cm^3
	Plastic Section Modulus, Z_{pz}	25100.25	cm^3
	Effective Slab Width, b_{eff}	2280	mm
	Transformed Composite I_z	4284758.2	cm^4
	Depth to Plastic Neutral Axis	262.8	mm

Continued on next page

Table 5.1: Girder Section Properties (Final Optimized / User-selected) (continued)

Girder	Property	Value	Unit
G3	Depth, D	1670	mm
	Top Flange Width, b_f	510	mm
	Bottom Flange Width, b_f	510	mm
	Top Flange Thickness, t_f	22	mm
	Bottom Flange Thickness, t_f	22	mm
	Web Thickness, t_w	10	mm
	Gross Area, A	387	cm^2
	Moment of Inertia, I_z	1881957.85	cm^4
	Elastic Section Modulus, Z_{ez}	22538.42	cm^3
	Plastic Section Modulus, Z_{pz}	25100.25	cm^3
	Effective Slab Width, b_{eff}	2280	mm
	Transformed Composite I_z	4284758.2	cm^4
	Depth to Plastic Neutral Axis	262.8	mm

Continued on next page

Table 5.1: Girder Section Properties (Final Optimized / User-selected) (continued)

Girder	Property	Value	Unit
G4	Depth, D	1670	mm
	Top Flange Width, b_f	510	mm
	Bottom Flange Width, b_f	510	mm
	Top Flange Thickness, t_f	22	mm
	Bottom Flange Thickness, t_f	22	mm
	Web Thickness, t_w	10	mm
	Gross Area, A	387	cm^2
	Moment of Inertia, I_z	1881957.85	cm^4
	Elastic Section Modulus, Z_{ez}	22538.42	cm^3
	Plastic Section Modulus, Z_{pz}	25100.25	cm^3
	Effective Slab Width, b_{eff}	2280	mm
	Transformed Composite I_z	4284758.2	cm^4
	Depth to Plastic Neutral Axis	262.8	mm
G5	Depth, D	1670	mm
	Top Flange Width, b_f	510	mm
	Bottom Flange Width, b_f	510	mm
	Top Flange Thickness, t_f	22	mm
	Bottom Flange Thickness, t_f	22	mm
	Web Thickness, t_w	10	mm
	Gross Area, A	387	cm^2
	Moment of Inertia, I_z	1881957.85	cm^4
	Elastic Section Modulus, Z_{ez}	22538.42	cm^3
	Plastic Section Modulus, Z_{pz}	25100.25	cm^3
	Effective Slab Width, b_{eff}	2280	mm
	Transformed Composite I_z	4284758.2	cm^4
	Depth to Plastic Neutral Axis	262.8	mm

Table 5.2: Girder Section Classification

Girder	Element	Slenderness Ratio	Class Limit	Classification
G1	Top Flange	11.36	11.49	Semi-Compact
	Web	162.6	106.49	Slender
	Overall Section	—	—	Slender
G2	Top Flange	11.36	11.49	Semi-Compact
	Web	162.6	106.49	Slender
	Overall Section	—	—	Slender
G3	Top Flange	11.36	11.49	Semi-Compact
	Web	162.6	106.49	Slender
	Overall Section	—	—	Slender
G4	Top Flange	11.36	11.49	Semi-Compact
	Web	162.6	106.49	Slender
	Overall Section	—	—	Slender
G5	Top Flange	11.36	11.49	Semi-Compact
	Web	162.6	106.49	Slender
	Overall Section	—	—	Slender

IS 800:2007 Table 2.

Table 5.3: Moment Capacity Check

Girder	Parameter	Formula / Reference	Value	Status
G1	Applied Moment, M_u	Governing LC (ULS)	2646.84 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	11477.22 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.23	**PASS**
G2	Applied Moment, M_u	Governing LC (ULS)	2646.84 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	11477.22 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.23	**PASS**
G3	Applied Moment, M_u	Governing LC (ULS)	2646.84 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	11477.22 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.23	**PASS**

Continued on next page

Table 5.3: Moment Capacity Check (continued)

Girder	Parameter	Formula / Reference	Value	Status
G4	Applied Moment, M_u	Governing LC (ULS)	2646.84 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	11477.22 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.23	**PASS**
G5	Applied Moment, M_u	Governing LC (ULS)	2646.84 kN-m	—
	Design Moment Capacity, M_d	IRC 22 Cl. 603.3.1	11477.22 kN-m	—
	Utilization Ratio, M_u/M_d	M_u/M_d	0.23	**PASS**

IRC 22 Cl. 603.3.1, IS 800 Cl. 8.2.1.

Table 5.4: Shear Capacity Check

Girder	Parameter	Formula / Reference	Value	Status
G1	Applied Shear, V_u	Governing LC (ULS)	351.11 kN	—
	Shear Area, A_v	$d_w \times t_w$	16260 mm^2	—
	Panel Aspect Ratio, c/d	—	1.501	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.126	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	2.037	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.72 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	720.21 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.49	**PASS**
G2	Applied Shear, V_u	Governing LC (ULS)	351.11 kN	—
	Shear Area, A_v	$d_w \times t_w$	16260 mm^2	—
	Panel Aspect Ratio, c/d	—	1.501	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.126	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	2.037	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.72 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	720.21 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.49	**PASS**

Continued on next page

Table 5.4: Shear Capacity Check (continued)

Girder	Parameter	Formula / Reference	Value	Status
G3	Applied Shear, V_u	Governing LC (ULS)	351.11 kN	—
	Shear Area, A_v	$d_w \times t_w$	16260 mm^2	—
	Panel Aspect Ratio, c/d	—	1.501	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.126	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	2.037	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.72 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	720.21 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.49	**PASS**
G4	Applied Shear, V_u	Governing LC (ULS)	351.11 kN	—
	Shear Area, A_v	$d_w \times t_w$	16260 mm^2	—
	Panel Aspect Ratio, c/d	—	1.501	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.126	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	2.037	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.72 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	720.21 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.49	**PASS**
G5	Applied Shear, V_u	Governing LC (ULS)	351.11 kN	—
	Shear Area, A_v	$d_w \times t_w$	16260 mm^2	—
	Panel Aspect Ratio, c/d	—	1.501	—
	Shear Buckling Coefficient, k_v	IS 800 Cl. 8.4.2.2	7.126	—
	Web Slenderness, λ_w	$\sqrt{f_{yw}/(\sqrt{3}\,\tau_{cr})}$	2.037	—
	Design Shear Stress, τ_b	IRC 22 Cl. 603.3.3.2	48.72 MPa	—
	Shear Buckling Resistance, V_{cr}	$A_v \times \tau_b$	720.21 kN	—
	Utilization Ratio, V_u/V_d	V_u/V_d	0.49	**PASS**

IS 800 Cl. 8.4, IRC 22 Cl. 603.3.3.2.

Table 5.5: Interaction Checks (M-V and M-N)

Girder	Check	Condition	Value	Status
G1	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	11477.22 kN-m	—
	Interaction: $M_u \le M_{dv}$	M_u/M_{dv}	0.23	**PASS**
	Interaction: $N_u/N_{Rd} + M_u/M_{dv} \le 1.0$	$0.00 + 0.23 = 0.234$	0.234	**PASS**
G2	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	11477.22 kN-m	—
	Interaction: $M_u \le M_{dv}$	M_u/M_{dv}	0.23	**PASS**
	Interaction: $N_u/N_{Rd} + M_u/M_{dv} \le 1.0$	$0.00 + 0.23 = 0.234$	0.234	**PASS**
G3	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	11477.22 kN-m	—
	Interaction: $M_u \le M_{dv}$	M_u/M_{dv}	0.23	**PASS**
	Interaction: $N_u/N_{Rd} + M_u/M_{dv} \le 1.0$	$0.00 + 0.23 = 0.234$	0.234	**PASS**
G4	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	11477.22 kN-m	—
	Interaction: $M_u \le M_{dv}$	M_u/M_{dv}	0.23	**PASS**
	Interaction: $N_u/N_{Rd} + M_u/M_{dv} \le 1.0$	$0.00 + 0.23 = 0.234$	0.234	**PASS**
G5	High Shear Condition?	$V_u > 0.6\,V_d$	No	—
	Reduced Moment Capacity, M_{dv}	IRC 22 Cl. 603.3.3.3	11477.22 kN-m	—
	Interaction: $M_u \le M_{dv}$	M_u/M_{dv}	0.23	**PASS**
	Interaction: $N_u/N_{Rd} + M_u/M_{dv} \le 1.0$	$0.00 + 0.23 = 0.234$	0.234	**PASS**

IRC 22 Cl. 603.3.3.3.

Table 5.6: Lateral Torsional Buckling Check — Construction Stage

Girder	Parameter	Formula / Reference	Value	Status
G1	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	88107.8 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.299	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	7986.44 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.24	**PASS**
G2	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	88107.8 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.299	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	7986.44 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.24	**PASS**
G3	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	88107.8 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.299	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	7986.44 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.24	**PASS**
G4	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	88107.8 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.299	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	7986.44 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.24	**PASS**
G5	Elastic Critical Moment, M_{cr}	IRC 22 Cl. 603.3.3.1	88107.8 kN-m	—
	Non-dim. Slenderness, $\bar{\lambda}_{LT}$	$\sqrt{M_p/M_{cr}}$	0.299	—
	LTB Reduction Factor, χ_{LT}	IS 800 Cl. 8.2.2	1	—
	LTB Resistance, M_b	$\chi_{LT}\, M_p/\gamma_{m0}$	7986.44 kN-m	—
	$M_u \leq M_b$	M_u/M_b	0.24	**PASS**

IRC 22 Cl. 603.3.3.1, IS 800 Cl. 8.2.2.

Table 5.7: Stiffener Design Summary

Girder	Parameter	Value	Unit
G1	Shear Buckling Design Method	post_critical	—
	Intermediate Stiffener Thickness	21.1	mm
	Intermediate Stiffener Spacing	—	mm
	End Panel Stiffener Thickness	1.5	mm
	No. of End Panel Stiffeners	4	—
	Longitudinal Stiffeners	None	—
G2	Shear Buckling Design Method	post_critical	—
	Intermediate Stiffener Thickness	21.1	mm
	Intermediate Stiffener Spacing	—	mm
	End Panel Stiffener Thickness	1.5	mm
	No. of End Panel Stiffeners	4	—
	Longitudinal Stiffeners	None	—
G3	Shear Buckling Design Method	post_critical	—
	Intermediate Stiffener Thickness	21.1	mm
	Intermediate Stiffener Spacing	—	mm
	End Panel Stiffener Thickness	1.5	mm
	No. of End Panel Stiffeners	4	—
	Longitudinal Stiffeners	None	—
G4	Shear Buckling Design Method	post_critical	—
	Intermediate Stiffener Thickness	21.1	mm
	Intermediate Stiffener Spacing	—	mm
	End Panel Stiffener Thickness	1.5	mm
	No. of End Panel Stiffeners	4	—
	Longitudinal Stiffeners	None	—
G5	Shear Buckling Design Method	post_critical	—
	Intermediate Stiffener Thickness	21.1	mm
	Intermediate Stiffener Spacing	—	mm
	End Panel Stiffener Thickness	1.5	mm
	No. of End Panel Stiffeners	4	—
	Longitudinal Stiffeners	None	—

Table 5.8: End Panel Stiffener Checks

Girder	Check	Required	Provided	Status
G1	Web Buckling Resistance	472.49 kN	168429.67 kN	**PASS**
	Local Crushing Resistance	472.49 kN	577.49 kN	**PASS**
	Bearing Capacity	472.49 kN	3181.82 kN	**PASS**
	Column Buckling Resistance	472.49 kN	2863.64 kN	**PASS**
G2	Web Buckling Resistance	472.49 kN	168429.67 kN	**PASS**
	Local Crushing Resistance	472.49 kN	577.49 kN	**PASS**
	Bearing Capacity	472.49 kN	3181.82 kN	**PASS**
	Column Buckling Resistance	472.49 kN	2863.64 kN	**PASS**
G3	Web Buckling Resistance	472.49 kN	168429.67 kN	**PASS**
	Local Crushing Resistance	472.49 kN	577.49 kN	**PASS**
	Bearing Capacity	472.49 kN	3181.82 kN	**PASS**
	Column Buckling Resistance	472.49 kN	2863.64 kN	**PASS**
G4	Web Buckling Resistance	472.49 kN	168429.67 kN	**PASS**
	Local Crushing Resistance	472.49 kN	577.49 kN	**PASS**
	Bearing Capacity	472.49 kN	3181.82 kN	**PASS**
	Column Buckling Resistance	472.49 kN	2863.64 kN	**PASS**
G5	Web Buckling Resistance	472.49 kN	168429.67 kN	**PASS**
	Local Crushing Resistance	472.49 kN	577.49 kN	**PASS**
	Bearing Capacity	472.49 kN	3181.82 kN	**PASS**
	Column Buckling Resistance	472.49 kN	2863.64 kN	**PASS**

IS 800 Cl. 8.7.3; required value = support reaction.

Table 5.9: Serviceability — Deflection Checks

Girder	Check	Allowable	Actual	Status
G1	Live Load Deflection, δ_{LL}	L/800 = 37.5 mm	24.948 mm	**PASS**
	Total Load Deflection, δ_{total}	L/600 = 50.0 mm	61.490 mm	**FAIL**
G2	Live Load Deflection, δ_{LL}	L/800 = 37.5 mm	25.097 mm	**PASS**
	Total Load Deflection, δ_{total}	L/600 = 50.0 mm	61.697 mm	**FAIL**
G3	Live Load Deflection, δ_{LL}	L/800 = 37.5 mm	25.150 mm	**PASS**
	Total Load Deflection, δ_{total}	L/600 = 50.0 mm	61.738 mm	**FAIL**
G4	Live Load Deflection, δ_{LL}	L/800 = 37.5 mm	25.033 mm	**PASS**
	Total Load Deflection, δ_{total}	L/600 = 50.0 mm	61.550 mm	**FAIL**
G5	Live Load Deflection, δ_{LL}	L/800 = 37.5 mm	24.737 mm	**PASS**
	Total Load Deflection, δ_{total}	L/600 = 50.0 mm	61.156 mm	**FAIL**

IRC 22 Cl. 604.3.2.

Table 5.10: Serviceability — Maximum Stress Limitation

Girder	Element	Allowable Stress	Actual Stress	Status
G1	Structural Steel ($0.9\,f_y$)	315.00 MPa	93.56 MPa	**PASS**
G2	Structural Steel ($0.9\,f_y$)	315.00 MPa	93.56 MPa	**PASS**
G3	Structural Steel ($0.9\,f_y$)	315.00 MPa	93.56 MPa	**PASS**
G4	Structural Steel ($0.9\,f_y$)	315.00 MPa	93.56 MPa	**PASS**
G5	Structural Steel ($0.9\,f_y$)	315.00 MPa	93.56 MPa	**PASS**

IRC 22 Cl. 604.3.1.

Table 5.11: Serviceability — Fatigue Assessment

Girder	Stress Range, $\Delta\sigma$ (MPa)	Fatigue Limit, f_{fd} (MPa)	Utilization Ratio	Status
G1	166.88	92.49	**1.80**	**FAIL**
G2	167.84	92.49	**1.81**	**FAIL**
G3	169.33	92.49	**1.83**	**FAIL**
G4	171.72	92.49	**1.86**	**FAIL**
G5	173.02	92.49	**1.87**	**FAIL**

IRC 22 Cl. 605 — governing of normal and shear fatigue (worst by DCR). Capacity reduction factor μ_r applied where plate thickness > 25 mm.

Table 5.12: Girder Design Summary (DCR / Utilization Ratio)

Girder	Controlling LC / Combination	Controlling Check	Demand	Capacity	UR	Status
G1	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	166.88 MPa	92.49 MPa	**1.804**	**FAIL**
G2	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	167.84 MPa	92.49 MPa	**1.815**	**FAIL**
G3	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	169.33 MPa	92.49 MPa	**1.831**	**FAIL**
G4	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	171.72 MPa	92.49 MPa	**1.857**	**FAIL**
G5	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	Fatigue Normal Stress	173.02 MPa	92.49 MPa	**1.871**	**FAIL**

UR = Demand / Capacity. A value ≤ 1.0 indicates a passing check. The controlling check is the criterion with the highest UR for each girder, with the real load case / combination that drives it.

Table 5.13: Shear Connector Capacity

Parameter	Formula	Value	Reference
Design Resistance, Q_u	$Q_u = \min(Q_{u,s},\, Q_{u,c})$ $Q_{u,s} = \dfrac{0.8\, f_u\, (\pi d^2/4)}{\gamma_v}$ $Q_{u,c} = \dfrac{0.29\, \alpha\, d^2 \sqrt{f_{ck}\, E_{cm}}}{\gamma_v}$	95.36 kN	IRC 22 Cl. 606.3.1 (Eq. 6.1)
Fatigue Shear Resistance, Q_r	IRC 22 Table 8 (ϕd, N_{sc})	25.00 kN	IRC 22 Cl. 606.3.2 (Table 8)

Table 5.14: Shear Connector Spacing

Criterion	Governing Spacing	Actual Spacing Provided	Status
ULS Shear (SL1)	1795.4 mm	348.6 mm	**PASS**
Full Composite (SL2)	348.6 mm	348.6 mm	**PASS**
SLS Fatigue (SR)	482.9 mm	348.6 mm	**PASS**
Max Spacing Limit (IRC 22)	400.0 mm	348.6 mm	**PASS**

IRC 22 Cl. 606.4, 606.9. Governing spacing $= \min(S_{L1}, S_{L2}, S_R)$.

Table 5.15: Transverse Shear and Detailing Checks

Check	Value	Status
Longitudinal Shear per unit length, V_L	106.23 kN/m	—
Transverse Shear Capacity of Slab, V_{Rd}	999.28 kN/m	—
Transverse Shear Check	$V_L/V_{Rd} = 0.11$	**PASS**
Min. Transverse Reinforcement, $A_{st,min}$	Required 0.53 cm^2/m, Provided 11.77 cm^2/m	**PASS**
Stud Diameter $\leq 2\,t_f$	$d = 20.0$ mm $\leq 2t_f = 44.0$ mm	**PASS**
Stud Edge Distance	Provided 195.0 mm (req. $\geq$ 25.0 mm)	**PASS**

IRC 22 Cl. 606.6, 606.10.

5.2 Deck Slab Design

The reinforced concrete deck slab is designed per IRC 112:2011 (flexure, shear, crack width) and IRC 22:2014 (composite construction). Wheel loads are distributed using Pigeaud's method. The deck is checked for flexure in the transverse and longitudinal directions, punching shear, one-way (beam) shear, crack width, and reinforcement detailing.

Table 5.16: Deck Slab — Loading and Geometry

Parameter	Value	Unit	Reference / Remarks
Effective Span of Deck Slab, l_{eff}	2280	mm	Girder spacing (c/c)
Deck Thickness, t_s	250	mm	Additional Inputs
Clear Cover	Top 50 / Bottom 50	mm	IRC 112 Cl. 15.2
Concrete Grade	M40	—	$f_{ck} = 40$ MPa, $f_{ctm} = 3$ MPa (IRC 112 Cl. 6.4)
Reinforcement Grade	Fe 500	—	$f_y = 500$ MPa (IRC 112 Cl. 6.2)
Dead Load per Unit Area, w_{DL}	6.13	kN/m^2	Slab self-weight
IRC 6 Wheel Load	83.4	kN	Governing wheel, IRC 6 Cl. 204
Tyre Contact Width	300	mm	Transverse, IRC 6 Annex A
Impact Factor	1.100	—	IRC 6 Cl. 208
Governing Live Load Case	Class70R(W)	—	Vehicle giving the largest moment

Table 5.17: Deck Slab — Flexure Check: Interior Panel (Pigeaud's Method)

Location	Parameter	Formula / Reference	Value	Status
At Midspan (Sagging)	Transverse BM (DL), $M_{T,DL}$	$w_{DL}\, l_{eff}^2/10$	3.19 kN-m/m	—
	Transverse BM (LL), $M_{T,LL}$	Effective width (IRC 112 B3.1)	26.77 kN-m/m	—
	Total Design BM, $M_{u,sag}$	1.35 DL + 1.50 LL	48.48 kN-m/m	—
	Effective depth, d	$t_s - c_{nom} - \phi/2$	194.0 mm	—
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	55.25 kN-m/m	**PASS**
At Support (Hogging)	Total Design BM, $M_{u,hog}$	1.35 DL + 1.50 LL (at support)	36.36 kN-m/m	—
	Required Top Steel, $A_{st,top}$	$M_u/(0.87\, f_y\, d)$	488 mm^2/m	—
	Moment Capacity, M_{Rd}	IRC 112 Cl. 12.2	40.16 kN-m/m	**PASS**

IRC 112 Cl. 12.2. Distribution (longitudinal) reinforcement designed for 20% of main steel moment (IRC 21 Cl. 305.18).

Table 5.18: Deck Slab — Cantilever Overhang Flexure Check

Parameter	Formula / Reference	Value	Status
Overhang Length, l_{oh}	—	1.14 m	—
Crash Barrier Load Moment	IRC 6 Cl. 206.4	7.50 kN-m/m	—
Dead Load Moment	$w_{DL}\, l_{oh}^2/2$ + railing	5.66 kN-m/m	—
Live Load Moment (eccentric wheel)	Wheel load × arm	51.98 kN-m/m	—
Total Hogging Moment, $M_{u,oh}$	1.35 DL + 1.50 (LL + CB)	104.67 kN-m/m	—
Moment Capacity (top steel), $M_{Rd,oh}$	IRC 112 Cl. 12.2	114.71 kN-m/m	**PASS**

IRC 6 Cl. 206.4 crash barrier loads applied at kerb face; IRC 112 Cl. 12.2 flexure.

Table 5.19: Deck Slab — Punching Shear Check (IRC 112 Cl. 10.4.6)

Parameter	Formula / Reference	Value	Status
Design Wheel Load (ULS), V_{Ed}	$\gamma_Q\,(1+IF)\,P_w$	137.6 kN	—
Tyre Contact Area	$a \times b$ (IRC 6 Annex A)	300 × 150 mm	—
Loaded Area at mid-depth, b_0	$c_1 \times c_2$ (incl. WC dispersion)	400 × 250 mm	—
Control Perimeter, u_1	$2(c_1+c_2)+4\pi d$	3738 mm	—
Punching Shear Stress, v_{Ed}	$V_{Ed}/(u_1\,d)$	0.190 MPa	—
Punching Resistance, $v_{Rd,c}$	IRC 112 Eq. 10.1	0.555 MPa	—
Punching Shear Check	$v_{Ed}/v_{Rd,c} \le 1.0$	0.34	**PASS**

Punching shear reinforcement not typically required for deck slabs with $d \ge 200$ mm and adequate longitudinal reinforcement.

Table 5.20: Crack Width Check (Deck Slab)

Parameter	Value	Unit	Reference / Remarks
Min. Reinforcement for Crack Control, $A_{s,min}$	303	mm^2/m	IRC 112 Cl. 16.5.1
Provided Reinforcement (bottom)	685	mm^2/m	ϕ12 @ 165 mm c/c
Max. Permissible Crack Width	0.30	mm	IRC 112 Table 12.1
Calculated Crack Width, w_k (governing)	0.2252	mm	Frequent SLS combination
Crack Width Check	**PASS**	—	$w_k \le w_{k,max}$

Table 5.21: One-Way (Beam) Shear Check (Deck Slab)

Parameter	Formula / Reference	Value	Status
Design Shear per unit width, V_{Ed}	$\gamma_{DL}V_{DL}+\gamma_{LL}(1+IF)V_{LL}$	86.94 kN/m	—
Effective depth, d	$t_s - c_{nom} - \phi/2$	194.0 mm	—
Size factor, k	$1+\sqrt{200/d} \le 2.0$	2.000	—
Long. reinforcement ratio, ρ_l	$A_{sl}/(b_w\,d) \le 0.02$	0.0035	—
Shear resistance (no stirrups), $V_{Rd,c}$	$v_{Rd,c}\,b_w\,d$ (Cl. 10.3.2)	107.58 kN/m	—
One-Way Shear Check	$V_{Ed}/V_{Rd,c} \le 1.0$	0.81	**PASS**

IRC 112 Cl. 10.3.2. Shear reinforcement not provided in deck slabs; capacity relies on concrete and main reinforcement.

Table 5.22: Reinforcement Detailing Summary (Deck Slab)

Parameter	Required / Limit	Provided	Status
Main Reinforcement — Bottom (Transverse)			
Required Area, $A_{st,req}$	631 mm^2/m	685 mm^2/m	**PASS**
Bar Diameter × Spacing	$\phi \geq 10$ mm (IRC 112)	ϕ12 @ 165 mm c/c	—
Min. Reinforcement $A_{s,min}$ (IRC 112 Cl. 16.3.1)	303 mm^2/m	685 mm^2/m	**PASS**
Max. Bar Spacing (IRC 112 Cl. 16.3.2)	300 mm	165 mm	**PASS**
Distribution Reinforcement — Longitudinal			
Required Area, $A_{st,dist}$	$\geq$ 20% of main steel	377 mm^2/m	**PASS**
Top Reinforcement (Support / Cantilever Overhang)			
Required Area, $A_{st,top}$	488 mm^2/m	492 mm^2/m	**PASS**
Cover and Detailing			
Clear Cover (IRC 112 Cl. 15.2)	$\geq$ 50 mm (Table 14.2)	Top 50 / Bottom 50 mm	**PASS**

IRC 112 Cl. 16.3, IS 456 Cl. 26.5.

5.3 Cross Bracing Design

Cross bracing between adjacent plate girders provides lateral stability during construction, resists transverse loads (wind, seismic, braking) in service, and prevents lateral torsional buckling of the girders. Members are designed per IS 800:2007 Cl. 7 (compression) and Cl. 6 (tension). Forces are derived from the grillage model under the governing load combination.

Table 5.23: Cross Bracing — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm²)	r_{min} (mm)
G1-G2	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G2-G3	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G3-G4	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G4-G5	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0

A_g = gross cross-sectional area; r_{min} = minimum radius of gyration.

Table 5.24: Cross Bracing — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit	Status
G1-G2	Diagonal	C	2686	141.9	250	**PASS**
	Top chord	C	2280	120.5	250	**PASS**
	Bottom chord	T	2280	120.5	400	**PASS**
G2-G3	Diagonal	C	2686	141.9	250	**PASS**
	Top chord	C	2280	120.5	250	**PASS**
	Bottom chord	T	2280	120.5	400	**PASS**
G3-G4	Diagonal	C	2686	141.9	250	**PASS**
	Top chord	C	2280	120.5	250	**PASS**
	Bottom chord	T	2280	120.5	400	**PASS**
G4-G5	Diagonal	C	2686	141.9	250	**PASS**
	Top chord	C	2280	120.5	250	**PASS**
	Bottom chord	T	2280	120.5	400	**PASS**

Limit = 250 for compression members (C), 400 for tension members (T). $K = 1.0$ for members with both ends pinned.

Table 5.25: Cross Bracing Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal	40 x 40 x 3	1.5 EQ (b)	10.09	15.35	0.93	**PASS**
	Chord	40 x 40 x 3	1.5 EQ (b)	8.57	19.71	0.79	**PASS**
G2-G3	Diagonal	40 x 40 x 3	1.5 EQ (a)	6.52	45.00	0.14	**PASS**
	Chord	40 x 40 x 3	1.5 EQ (a)	5.53	45.00	0.12	**PASS**
G3-G4	Diagonal	40 x 40 x 3	1.5 EQ (a)	7.18	45.00	0.16	**PASS**
	Chord	40 x 40 x 3	1.5 EQ (a)	6.10	45.00	0.14	**PASS**
G4-G5	Diagonal	40 x 40 x 3	1.5 EQ (b)	11.45	45.00	0.25	**PASS**
	Chord	40 x 40 x 3	1.5 EQ (b)	9.72	45.00	0.22	**PASS**

Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension) with the Osdag member-design module.

5.4 End Diaphragm Design

End diaphragms at the supports transfer transverse loads to the bearings, restrain the bottom flanges against lateral displacement, and maintain the girder cross-section geometry during construction and in service. They are designed per IS 800:2007 and IRC 24:2010 Cl. 507.

Table 5.26: End Diaphragm — Connection and Section Properties

Panel	Member	Connection	Section	A_g (mm^2)	r_{min} (mm)
G1-G2	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G2-G3	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G3-G4	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0
G4-G5	Diagonal	Bolted	40 x 40 x 3	237.0	8.0
	Top / Bottom chord	Bolted	40 x 40 x 3	237.0	8.0

A_g = gross cross-sectional area; r_{min} = minimum radius of gyration. The end diaphragms are cross-bracing frames; the analysis reports no separate end-diaphragm member forces, so the governing cross-bracing member design is shown.

Table 5.27: End Diaphragm — Slenderness Ratio Check (IS 800 Cl. 3.8)

Panel	Member	Nature	Eff. Length KL (mm)	KL/r	Limit	Status
G1-G2	Diagonal	C	2686	141.9	250	**PASS**
	Top chord	C	2280	120.5	250	**PASS**
	Bottom chord	T	2280	120.5	400	**PASS**
G2-G3	Diagonal	C	2686	141.9	250	**PASS**
	Top chord	C	2280	120.5	250	**PASS**
	Bottom chord	T	2280	120.5	400	**PASS**
G3-G4	Diagonal	C	2686	141.9	250	**PASS**
	Top chord	C	2280	120.5	250	**PASS**
	Bottom chord	T	2280	120.5	400	**PASS**
G4-G5	Diagonal	C	2686	141.9	250	**PASS**
	Top chord	C	2280	120.5	250	**PASS**
	Bottom chord	T	2280	120.5	400	**PASS**

Limit = 250 for compression members (C), 400 for tension members (T). $K = 1.0$ for members with both ends pinned.

Table 5.28: End Diaphragm Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal	40 x 40 x 3	1.5 EQ (b)	10.09	15.35	0.93	**PASS**
	Chord	40 x 40 x 3	1.5 EQ (b)	8.57	19.71	0.79	**PASS**
G2-G3	Diagonal	40 x 40 x 3	1.5 EQ (a)	6.52	45.00	0.14	**PASS**
	Chord	40 x 40 x 3	1.5 EQ (a)	5.53	45.00	0.12	**PASS**
G3-G4	Diagonal	40 x 40 x 3	1.5 EQ (a)	7.18	45.00	0.16	**PASS**
	Chord	40 x 40 x 3	1.5 EQ (a)	6.10	45.00	0.14	**PASS**
G4-G5	Diagonal	40 x 40 x 3	1.5 EQ (b)	11.45	45.00	0.25	**PASS**
	Chord	40 x 40 x 3	1.5 EQ (b)	9.72	45.00	0.22	**PASS**

Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension) with the Osdag member-design module. The end diaphragms are cross-bracing frames; the analysis reports no separate end-diaphragm member forces, so the governing cross-bracing member design is shown.

5.5 Overall Design Check Summary

The table below collects the governing design checks of all primary structural elements. The bar charts that follow show the utilization ratio (UR = demand / capacity) of each element; the red dashed line marks the limit UR = 1.0. **Steel Plate Girders** exceeds the limit and must be revised.

Table 5.29: Overall Design Check Summary — All Members

Element	Check	Governing Load Combination	Demand	Capacity	UR	Status
Steel Plate Girders	Moment	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	2646.84 kN-m	11477.22 kN-m	0.23	PASS
	Shear	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	472.49 kN	720.21 kN	0.66	PASS
	LTB (construction)	SW	220.66 kN-m	900.22 kN-m	0.25	PASS
	Deflection	1.0 DL + 1.0 LL	27.12 mm	50.00 mm	0.54	PASS
	Stress (SLS)	SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	97.15 MPa	315.00 MPa	0.31	PASS
	Fatigue	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	173.02 MPa	92.49 MPa	1.87	FAIL
Concrete Deck Slab	Flexure (sagging)	Basic ULS: 1.35DL + 1.5LL	48.48 kN-m/m	55.25 kN-m/m	0.88	PASS
	Flexure (hogging)	Basic ULS: 1.35DL + 1.5LL	36.36 kN-m/m	40.16 kN-m/m	0.91	PASS
	Cantilever overhang	Basic ULS: 1.35DL + 1.5LL	104.67 kN-m/m	114.71 kN-m/m	0.91	PASS
	Punching shear	Basic ULS: 1.35DL + 1.5LL	0.190 MPa	0.555 MPa	0.34	PASS
	One-way shear	Basic ULS: 1.35DL + 1.5LL	86.94 kN/m	107.58 kN/m	0.81	PASS
	Crack width	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	0.225 mm	0.300 mm	0.75	PASS
	Transverse shear (slab)	Envelope ULS	106.23 kN/m	999.28 kN/m	0.11	PASS
Cross Bracing	Compression	1.5 EQ (b)	14.28 kN	15.35 kN	0.93	PASS
	Tension	1.5 EQ (b)	11.25 kN	45.00 kN	0.25	PASS
	Slenderness	—	141.9	250.0	0.57	PASS
End Diaphragms	Compression	1.5 EQ (b)	14.28 kN	15.35 kN	0.93	PASS
	Tension	1.5 EQ (b)	11.25 kN	45.00 kN	0.25	PASS
	Slenderness	—	141.9	250.0	0.57	PASS

UR = Demand / Capacity. All values ≤ 1.0 indicate passing checks; values above 1.0 are highlighted. For each element the worst girder / panel / member is reported.

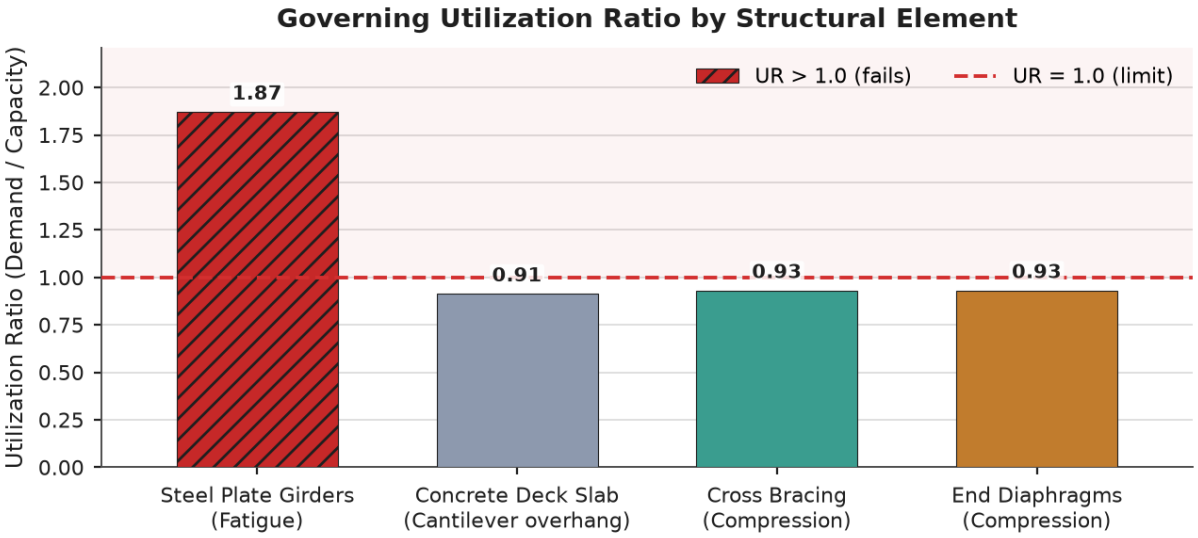


Figure 5.1: Governing utilization ratio of the primary structural elements (limit UR = 1.0)

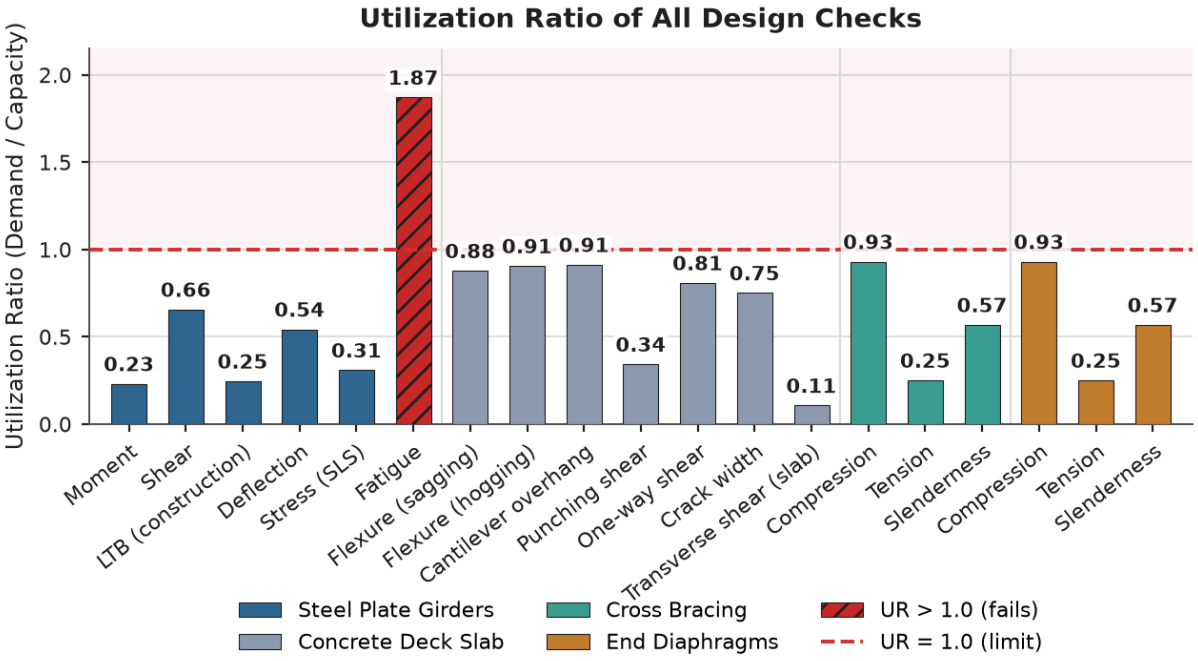


Figure 5.2: Utilization ratio of all summary design checks, grouped by structural element (limit UR = 1.0)

6 Drawings and Visualizations

This section presents CAD-generated views of the designed bridge and its components. All views are generated automatically by OsdagBridge using pythonOCC.

6.1 Bridge Configuration and Layout

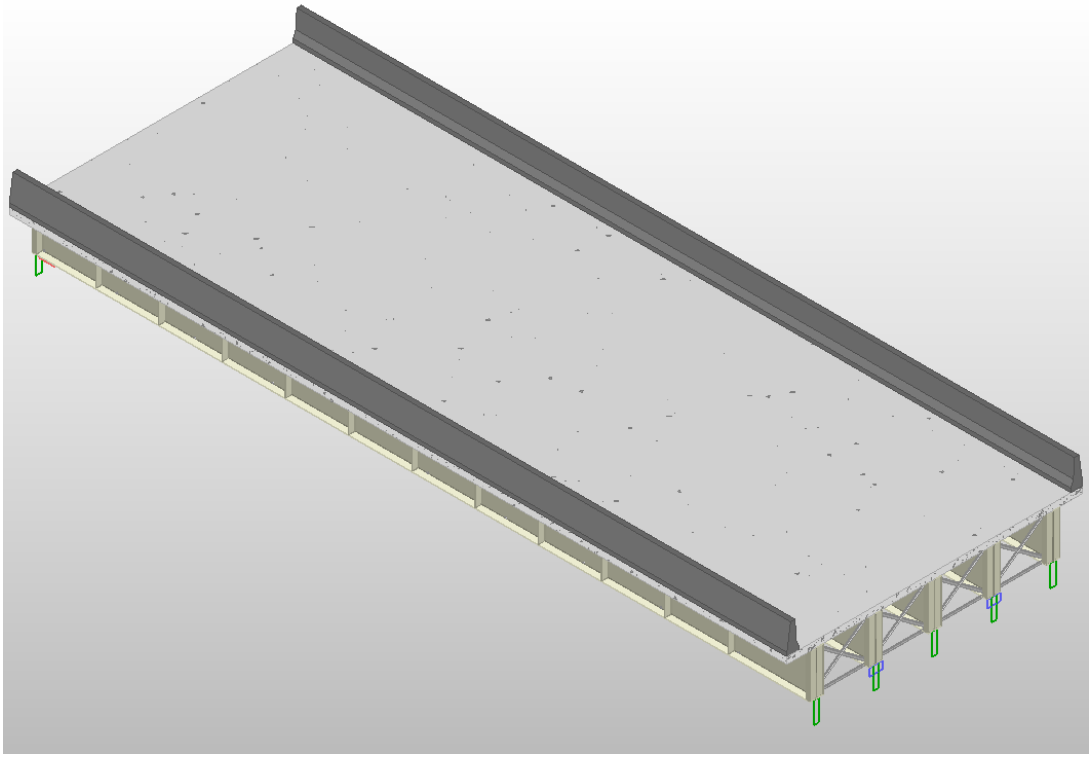


Figure 6.1: Overall 3D Bridge Superstructure

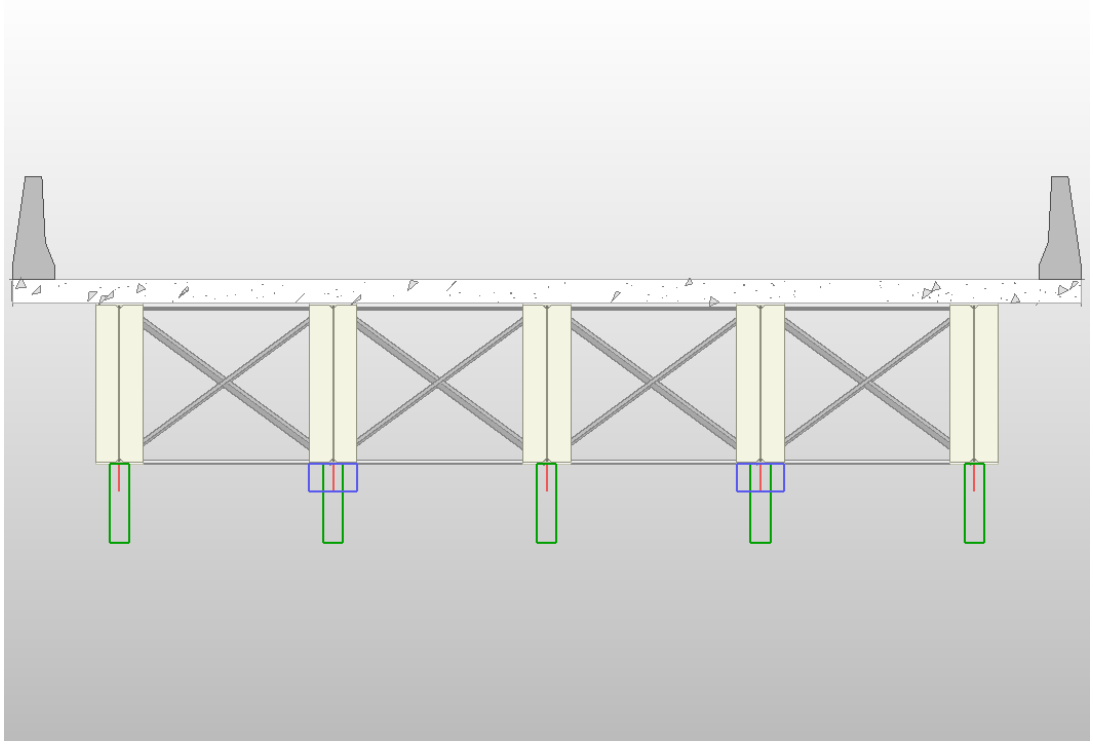


Figure 6.2: Typical Cross Section

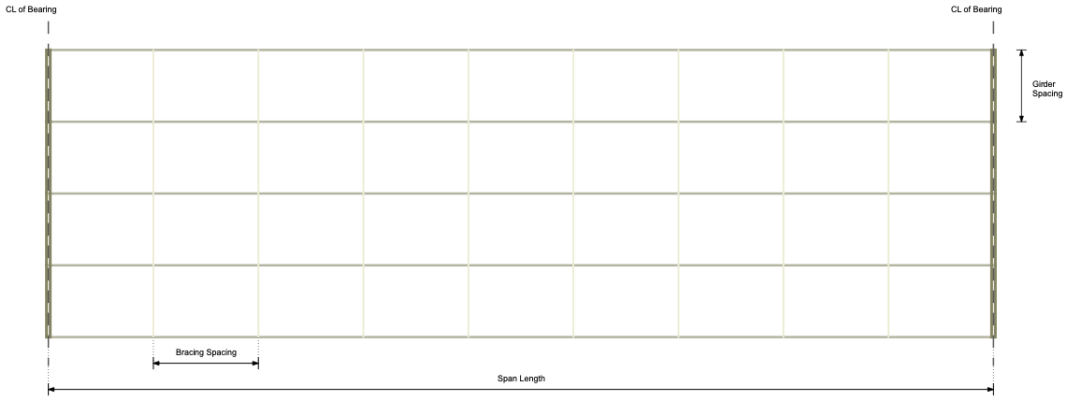


Figure 6.3: Top View

6.2 Plate Girder — Detailed Views

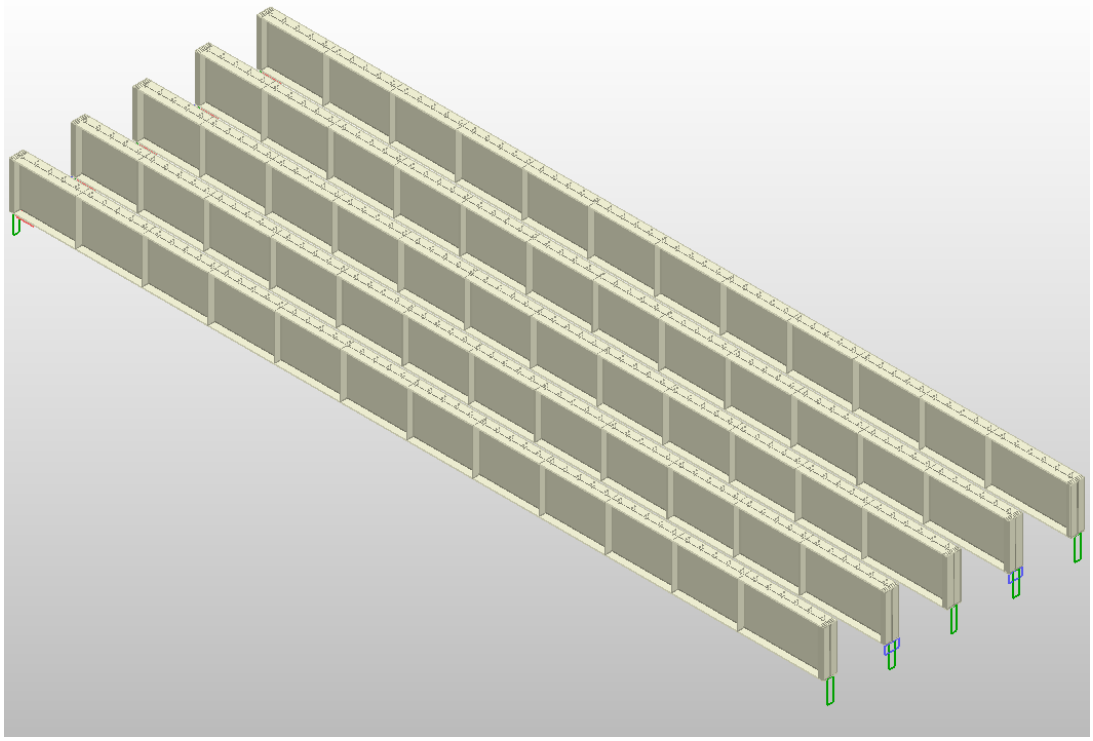


Figure 6.4: 3D View of Plate Girders

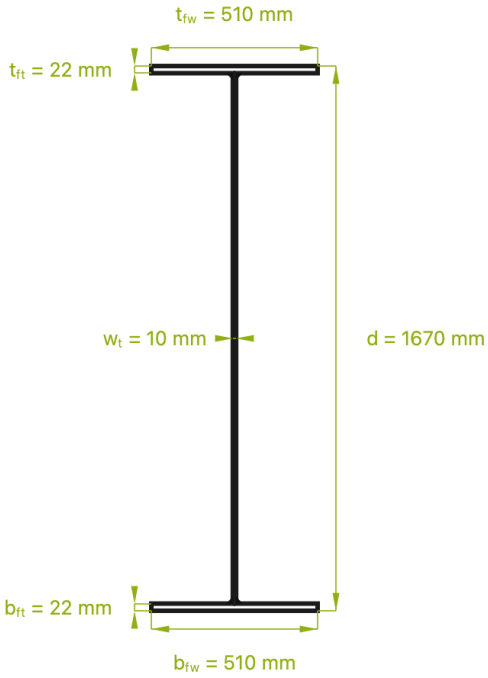


Figure 6.5: Cross Section of Plate Girder



Figure 6.6: Side View of Girder

6.3 Cross Bracing Detail

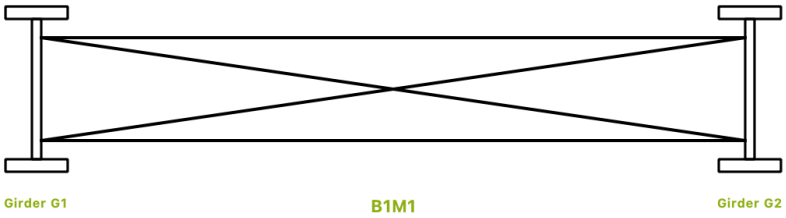


Figure 6.7: Cross Bracing Layout

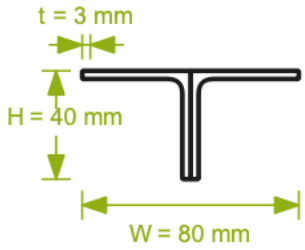


Figure 6.8: Cross Bracing – Bracing Section

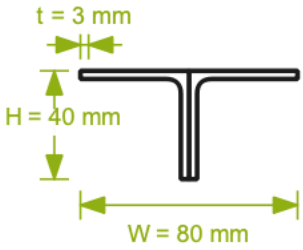


Figure 6.9: Cross Bracing – Top Chord Section

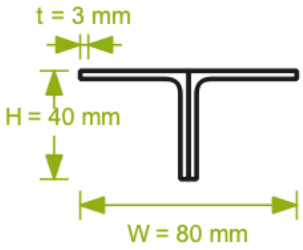


Figure 6.10: Cross Bracing – Bottom Chord Section

6.4 End Diaphragm Detail

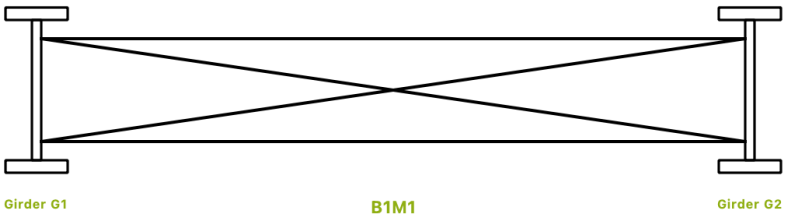


Figure 6.11: End Diaphragm Layout

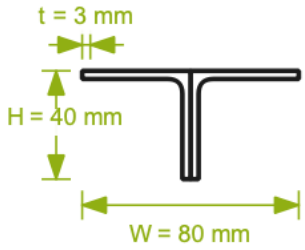


Figure 6.12: End Diaphragm – Bracing Section

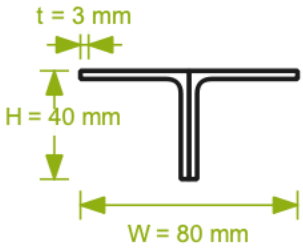


Figure 6.13: End Diaphragm – Top Chord Section

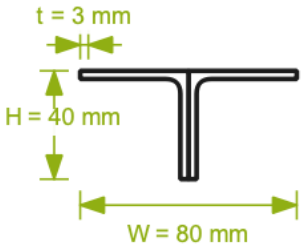


Figure 6.14: End Diaphragm – Bottom Chord Section

7 Material Take-off & Quantity Summary

This chapter lists the quantities of structural steel, concrete and reinforcement of the superstructure. The bill of materials is followed by a summary per material and by charts that show how the quantities are distributed.

7.1 Bill of Materials

Table 7.1: Bill of Materials (Steel, Concrete and Reinforcement Quantities)

S.N.	Item Description	Volume of One Item	Qty.	Total Volume (m^3)	Weight of One (MT)	Total Weight (MT)
1	Structural Steel (IS 2062) for Girders	0.03492 m^2 × 30.00 m = 1.04756 m^3	5	5.24	8.22	41.12
2(a)	Cross Bracing — Top Chord	0.00024 m^2 × 2.28 m = 0.00054 m^3	36	0.02	0.0042	0.15
2(b)	Cross Bracing — Bottom Chord	0.00024 m^2 × 2.28 m = 0.00054 m^3	36	0.02	0.0042	0.15
2(c)	Cross Bracing — Diagonals	0.00024 m^2 × 2.69 m = 0.00064 m^3	72	0.05	0.0050	0.36
3	End Diaphragms (frames at both supports)	0.00191 m^2 × 9.93 m = 0.01897 m^3	8	0.15	0.1489	1.19
4	Concrete for Deck Slab	11.40 m × 0.25 m × 30.00 m = 85.50 m^3	1	85.50	213.75	213.75
5	Reinforcement Steel for Deck Slab	0.043567 m^2 × 30.00 m = 1.30701 m^3	1	1.31	10.26	10.26
6	Shear Stud Connectors	0.000314 m^2 × 0.100 m = 0.000031 m^3	870	0.03	0.000247	0.215
7	Crash Barrier	0.30228 m^2 × 30.00 m = 9.06848 m^3	2	18.14	22.67	45.34

Unit weights: structural and reinforcement steel 7.85 MT/m^3, concrete 2.50 MT/m^3. Quantities are net (no allowance for wastage, splices, laps or connection plates).

Table 7.2: Quantity Summary by Material

Material / Component	Quantity	Unit	Share of Structural Steel
Structural steel			
Plate Girders	41.12	MT	95.7 %
Cross Bracing (chords + diagonals)	0.66	MT	1.5 %
End Diaphragms	1.19	MT	2.8 %
Total Structural Steel	42.97	MT	100.0 %
Deck slab			
Concrete Volume	85.50	m^3	—
Reinforcement Steel	10.26	MT	—
Reinforcement Content	120	kg/m^3	—

7.2 Material Quantity Charts

The charts below are generated from the quantities of the bill of materials.

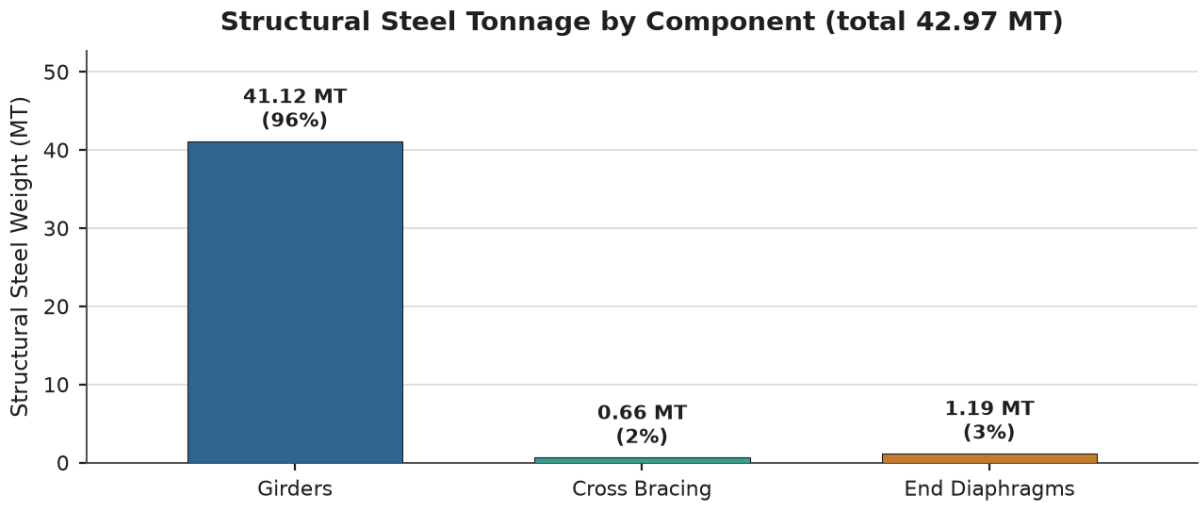


Figure 7.1: Structural steel tonnage by component (girders, cross bracing, end diaphragms)

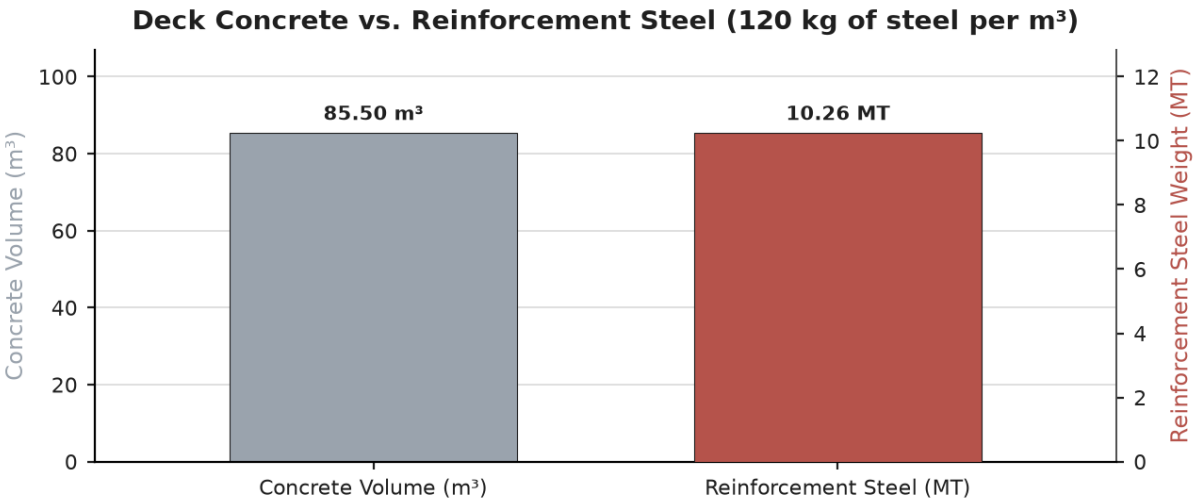


Figure 7.2: Deck slab: concrete volume (m^3) compared with reinforcement steel weight (MT)

8 Standards & Assumptions

This section provides references to standards used to calibrate the OsdagBridge design modules, and notes the limitations of the current software version.

8.1 Design Standards

The following Indian Road Congress (IRC) codes and Indian Standards (IS) form the basis of all design calculations in this software.

Table 8.1: IRC Codes

Code	Year	Title / Scope
IRC 5	2015	General Features of Design — carriageway widths, kerb, footpath dimensions
IRC 6	2017	Loads and Load Combinations — dead load, live load, impact, wind, temperature, etc.
IRC 22	2015	Composite Construction (LS) — composite section properties, ULS/SLS design, shear connectors
IRC 24	2010	Steel Road Bridges (LS) — stiffener design, skew angle limits, diaphragm requirements
IRC 112	2020	Concrete Road Bridges — deck slab flexure, shear, crack width, reinforcement
IRC SP 114	2018	Seismic Design of Road Bridges

Table 8.2: IS Codes

Code	Year	Scope
IS 800	2007	Steel construction — tension, compression, bending, shear, LTB, stiffeners, combined checks
IS 456	2000	Concrete — simplified stress-block for deck moment capacity
IS 1786	2008	Reinforcement steel properties
IS 1893 (Part 3)	2014	Earthquake resistant design
IS 2062	2011	Structural steel — yield and ultimate strength by grade

8.2 Analysis and Design Assumptions of This Version

Structural Analysis

- All girders are modelled as simply supported; continuous spans are not currently supported.
- A 3D grillage model (OSPGrillage) is used for load distribution. Grillage members carry composite section properties after the construction stage.
- The transverse member forces are computed using an approximate 2D frame analogy. For irregular or skewed geometries, a 3D FEM is recommended.
- Fixed bearing stiffness is modelled as $k = 1{,}000{,}000\,\text{kN/m}$ (virtually rigid); free/expansion bearing as $k = 100\,\text{kN/m}$.
- Construction stage sequence analysis is approximate. Detailed staged analysis should be performed for long-term deflection checks.

Composite Action

- Full shear connection is assumed at ULS with headed stud connectors designed per IRC 22:2015 Cl.606.
- Short-term composite section properties (modular ratio $n = E_s/E_{cm}$) are used for ULS checks.
- Long-term composite section properties (with creep-adjusted modular ratio) are used for SLS deflection and crack-width checks.
- Pre-composite stage: the steel girder alone resists all construction loads prior to concrete gaining strength.

Material Properties (IRC 22:2015 Annex III)

- Steel: $E_s = 200{,}000\,\text{MPa},\ G_s = 80{,}000\,\text{MPa},\ \nu = 0.30,\ \alpha = 11.7 \times 10^{-6}/°\text{C}$ (grade-independent).
- Minimum structural concrete grade: M25.
- Default reinforcement grade: Fe500.

Partial Safety Factors (IRC 22:2015 Cl.601.4)

- γ_{m0} (steel yield, ULS) = 1.10
- γ_{m1} (steel ultimate, ULS) = 1.25
- Reinforcement (ULS) = 1.15
- Welds – shop: 1.25; field: 1.50
- Fatigue (γ_{mft}) = 1.35

Loading

- Dead load densities per IRC 6:2017 Cl.203: structural steel 78.5 kN/m^3, concrete 25.0 kN/m^3, bituminous wearing course 24.0 kN/m^3.
- Multi-lane live load reduction factors per IRC 6:2017 Cl.204.4 Table 6A: 1st lane = 1.0, 2nd lane = 0.8, 3rd lane onwards = 0.4.
- Impact factor (dynamic load allowance) computed from span per IRC 6:2017 Cl.208.2/208.3.
- Wind load applied as transverse, longitudinal, and vertical components per IRC 6:2017 Cl.209.3.3–209.3.5.

Serviceability Limits

- Deflection limits (IRC 22:2015 Cl.604.3.2): live load + impact $\leq L/800$; total $\leq L/600$.
- SLS stress limits: concrete $\sigma_c \leq 0.48 f_{ck}$; rebar $\sigma_s \leq 0.80 f_{yk}$; steel $f_e \leq 0.9 f_y$.
- Permissible crack width: $w_k \leq 0.3$ mm (bridge deck, exposure class XS2/XD2 per IRC 112:2020 Cl.12.3.2).

Fatigue

- Reference fatigue life: $N_{sc} = 2 \times 10^6$ cycles (IRC 22:2015 Cl.605).
- Constant stress range is assumed. A thickness correction factor μ_r is applied for plate thickness > 25 mm.

Stiffener Design

- Intermediate transverse stiffener and bearing stiffener design follows IS 800:2007 Cl.8.7.2/8.7.3 and IRC 24:2010 Cl.509.7.2/509.7.3.

8.3 Known Limitations of This Version

- Substructure (piers, pile caps, foundations) and bearing design are not included.
- Splice connection design is not implemented.
- Skew angle > 15 degrees requires independent manual analysis (IRC 24 Cl. 504.8).
- Construction stage sequence analysis is approximate; detailed staged analysis should be performed for long-term deflection checks.
- The grillage analysis assumes simply supported boundary conditions; continuous spans are not currently supported.

References

1. IRC 5 (2024) — *Standard Specifications and Code of Practice for Road Bridges, Section I: General Features of Design.*
2. IRC 6 (2017) — *Standard Specifications and Code of Practice for Road Bridges, Section II: Loads and Load Combinations.*
3. IRC 22 (2014) — *Standard Specifications and Code of Practice for Road Bridges, Section VI: Composite Construction (Limit State Design).*
4. IRC 24 (2010) — *Standard Specifications and Code of Practice for Road Bridges, Section V: Steel Road Bridges (Limit State Method).*
5. IRC 112 (2011) — *Code of Practice for Concrete Road Bridges.*
6. IRC SP 114 (2018) — *Guidelines for Seismic Design of Road Bridges.*
7. IS 800 (2007) — *Indian Standard: General Construction in Steel — Code of Practice.*
8. IS 2062 (2011) — *Hot Rolled Medium and High Tensile Structural Steel — Specification.*
9. Subramanian, N. (2008). *Design of Steel Structures.* Oxford University Press.
10. Steel-INSDAG Teaching Resource Materials. `https://www.steel-insdag.org`